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**Develop models to estimate the lifetime value of customers based on usage patterns and
demographics**

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Declaration

I, Siri Chandana Dhurjati, confirm that this dissertation and the work presented in it are my own achievement. Where I have consulted the published work of others this is always clearly attributed; Where I have quoted from the work of others the source is always given. With the exception of such quotations this dissertation is entirely my own work; I have acknowledged all main sources of help; I have read and understand the penalties associated with Academic Misconduct.

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Abstract

In this study, machine learning techniques have been used to estimate Customer Lifetime Value (CLTV) in light of the limitations of the existing prediction methods. The research incorporated demographic and usage pattern data in order to enhance the accuracy of CLTV and optimize marketing, resource allocation, and customer relationship strategies. As a means for evaluation, the study uses 5 regression models (Linear Regression, Decision Tree, Random Forest, AdaBoost and XGBoost) using several performance metrics, such as RMSE, MAE and R^2 . The results highlight XGBoost as the most accurate model, able to fit a non-linear combination of factors and minimize errors.

The results illustrate how advanced predictive models help deal with problems such as data incoherency and dynamic customer behaviour. To achieve more refined predictions, they recommend more ensemble models, adding real-time data and adding more features. Based on this research, businesses are equipped with actionable insight to improve customer retention, target marketing efforts, and grow in a competitive market.

List of Abbreviations

- **AI:** Artificial Intelligence
- **CLTV:** Customer Lifetime Value
- **CRISP-DM:** Cross-Industry Standard Process for Data Mining
- **MAE:** Mean Absolute Error
- **MSE:** Mean Squared Error
- **R²:** Coefficient of Determination
- **RMSE:** Root Mean Squared Error
- **AWS:** Amazon Web Services
- **CRM:** Customer Relationship Management
- **CDP:** Customer Data Platform
- **RFM:** Recency, Frequency, Monetary Model

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Chapter 1: Introduction

1.1 Background of the study

Machine learning is used to determine Customer Lifetime Value (CLTV) by analysing demographic and usage trends. Companies value CLTV because it predicts a customer's lifetime revenue. This study will examine how customers' thoughts affect CLTV parameters, it will examine how customer behaviour and interactions effect long-term corporate performance, customer involvement, and lifetime value. Thus, the project tries to overcome CLTV estimation issues such data shortages and inaccurate estimations by creating models that blend demographic data with user behaviour patterns. The findings will improve CLTV prediction models. This will help businesses boost marketing, retain consumers, and maximise resources. The study will improve CLTV calculations and customer relationship management decisions using machine learning.

1.2 Rationale of the study

Businesses must understand their Customer Lifetime Value (CLTV) to improve marketing and budget allocation. The study is justified by the fact that typical CLTV estimation methods sometimes ignore client behaviour and current purchases. This study uses demographic and usage trend data and machine learning to improve CLTV predictions. As it helps organisations precisely estimate CLTV, this study helps them enhance customer relationship management and gain money over time. This can boost customer retention, targeted marketing, and business growth.

1.3 Problem Statement

Businesses must know how much a customer will spend on their items over time, this is especially true for organisations that wish to maintain good customer relations and better target their marketing mix to their greatest potential clients (Sifa *et al.* 2018). Most organisations struggle to calculate CLTV now. According to Segarra-Moliner *et al.* (2024) claim this leads to less marketing use, poor consumer communication, and low revenues. Most of these blunder's result from using old purchase data, which doesn't account for consumer interactions and behaviour.

This research aims to overcome these difficulties by constructing more complicated models using demographic data and customer behaviour trends. Learning more about clients will help develop more accurate CLTV estimates. Past CLTV predictions have likewise been flawed. Some essential criteria don't seem to account for changes in customers' personal or corporate circumstances or conduct that could affect future commercial interactions. These models don't consider environmental variables or how economic pressures like market fluctuations and customer purchasing power can affect behaviour.

The new predictive algorithms look at transactional records, customer attributes like age, geography, and income level, and behaviour indications like product use, satisfaction, feedback, etc. When combined, these models can better predict how the following factors may affect CLTV. These models will be trained using machine learning and data analysis so firms can make real-time changes depending on customer and market data.

This will likely develop into a collection of valuable tools that can aid organisations with marketing communications and customer management, notably keeping good customers and ensuring service satisfaction. This strategy is considered the greatest way to maximise resource use and better identify customers, which increases customer satisfaction. Building client loyalty with this Project will help businesses compete in public spaces and succeed in the long run.

1.4 Research Aim

The aim of the research is to estimate Customer Lifetime Value (CLTV) Using Demographic and Usage Pattern Analysis by The Application of Machine Learning Techniques.

1.5 Research Objectives

- To understand notions influencing customer perceptions and CLTV predictions.
- To explore relationships between customer engagement and lifetime value.
- To develop models that integrate demographics with user behaviour data.
- To identify challenges in current CLTV estimation methodologies.
- To propose solutions enhancing accuracy in CLTV forecasting models.

1.6 Research Questions

How can demographic and usage pattern data be leveraged to improve the accuracy and reliability of customer lifetime value predictions compared to traditional models?

1.7 Research Hypothesis

- H0: Integrating demographic data and usage patterns does not affect CLTV prediction accuracy.
- H1: Integrating demographic data and usage patterns improves CLTV prediction accuracy.

1.7 Significance of the research

The significance of this research will be to meet the project, first, the following should be included in a research report: study procedures, synthesised models, overviews and assessments, and conclusions. The main goal of this study is to describe and record the whole research process and its results. Second, this work will create a forecast model that can be used in real life to figure out how much a customer is worth whenever it's needed. This model will be useful in real life as well as scientifically sound. Finally, the new ideas and study results will be shared so that people can understand how the business and academic worlds work. These talks will be about how the Project works from a business point of view and how it might affect customer data.

Chapter 2: Literature Review

2.1 Introduction

This literature review will look at the current approaches and models used to figure out Customer Lifetime Value (CLTV), focussing on important ideas, problems, and new trends. It will look at how demographic information and patterns of customer behaviour are used in CLTV models, how machine learning can help make predictions better, and the problems businesses face when they try to record changing contacts with customers.

2.2 Development of theoretical framework

2.2.1 Concept of influence customer perceptions and CLTV predictions.

The companies that want to improve their marketing aspects, for them, “Customer Lifetime Value” has become one of the important measures as it allows the companies to keep more customers, and make more money (Kumari *et al.* 2024). The main purpose of integrating CLTV is to predict how much money a business can expect to make from a single customer over the period of their relationship.

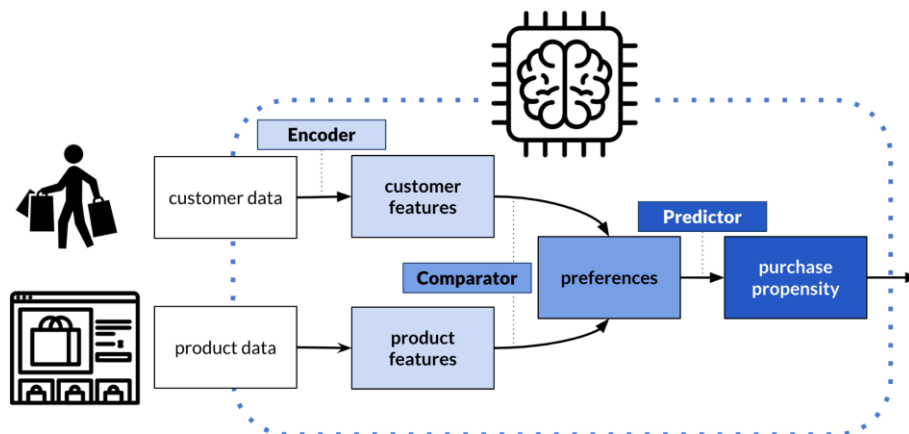


Figure 1 CLTV Predictions

(Source: Lahunou, 2021)

Demographic data such as age, income, area, and gender provide a fundamental understanding of what customers prefer. Thus, based on these factors, businesses can predict how much long-term value each customer adds by looking at these factors along with how often, what, and how

they use the channels of each customer. The models of machine learning, use a combination of data that forecast CLTV, providing the businesses with the opportunity to target high-value customers by sending them personalised messages or information (Ling *et al.* 2024). Therefore, by anticipating how certain demographic segments interact with the brand, companies can build trust and drive repeat business.

In order to understand this, here is an example of a company i.e., Amazon, which is using machine learning and demographic data to predict CLTV and influence the behaviour of customers (Marr, 2021).

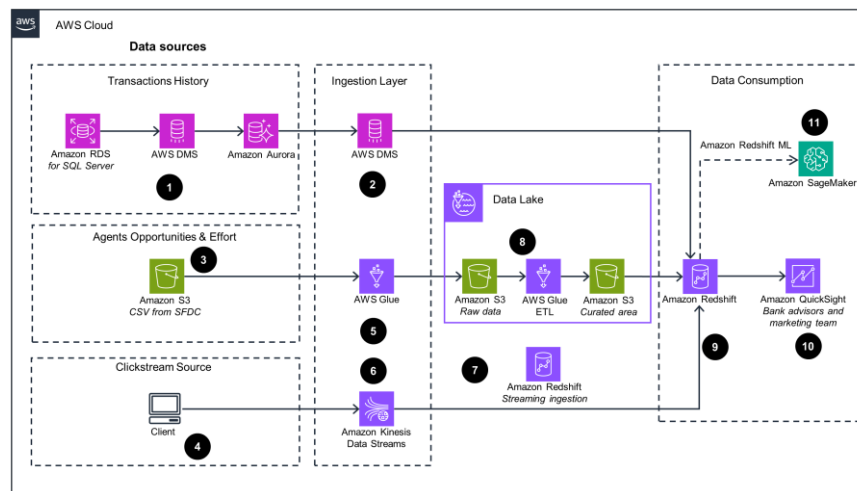


Figure 2 Guidance for customer lifetime value analysis on AWS

(Source: AWS, 2024)

Thus, to make better product suggestions, the company keeps track of what buyers look at, what they buy, and even their demographic information. Amazon uses algorithms to figure out which customers are most likely to buy from them again based on how they normally shop. This can change how those customers see the site as a place where they can get everything they need (Amazon Science, 2019). As a result, this improves the chances of repeat purchases by always providing suggestions that are relevant.

The concept of Customer Lifetime Value (CLTV) is derived from a marketing theory, which is considered a key parameter for measuring how profitable customer ties will be in the long run (Caldwell, 2022). One theory that a company uses in customer valuation for CLTV is “economic theory”. The theory demonstrates that customers add different amounts of value to

a business, over a period of time, and the factors which are considered are how much customers buy, how loyal they are, and how profitable they are (Kumar and Rajan, 2020). In addition to this, from the strategic marketing point of view, CLTV is considered by the company as it helps the company to come up with good strategies for getting new customers, keeping the one intact as well. Other than this, relationship marketing theory, stresses how important is to build long-lasting, mutually beneficial relationships with customers through personalised interactions, trust-building, and co-creating value (Dandis *et al.* 2022). It is also possible to understand the financial effects of CLTV on an organization's success by using financial theories. The Market-Based View (MBV) of the firm looks at how the market and how investors see the company affect its value.

2.2.2 Relationship between customer engagement and customer lifetime value.

According to Loureiro *et al.* (2020), the role of CEB in value co-creation is "the customer providing resources during non-transactional, combined value processes that happen in interaction with the focal firm and/or other stakeholders, thus affecting their individual value processes and outcomes." As a result, customer participation helps manufacturers speed up the product development process and come up with new products (Vogel *et al.* 2023).

Customer engagement and Customer Lifetime Value (CLTV) are two important measures that affect how profitable a business will be in the long run (Tripathi *et al.* 2021). When a customer keeps interacting with a brand through different channels, like social media, email, apps, and in-store events, that's called customer engagement. On the other hand, CLTV is the expected net profit that a business can expect from a customer over the course of their partnership (Chellaboina and Gembali, 2022). Businesses that involve their customers well build stronger relationships with them, make them more loyal to the brand, and get them to buy from them again and again, which all leads to a higher CLTV. Customers are more likely to be engaged and respect a brand when they interact with it more. This primarily happens because of tailoring experiences according to each customer and adapting to their changing needs. In today's competitive market, businesses are focussing more and more on boosting customer engagement to boost CLTV.

a) Personalisation:

One of the main things that keep customers interested is personalised interactions and experiences, which have a direct effect on CLTV. Companies build a stronger relationship with their customers when they use data analytics and machine learning to make personalised promotions, product ideas, and recommendations (Miklosik and Evans, 2020). *For example*, Amazon was one of the first companies to personalise the shopping experience of its customers. In order to make personalised suggestions, the company looks at a customer's past purchases, browsing habits, and demographic information (Lindecrantz *et al.* 2020). Amazon increases CLTV by getting customers more involved with the shopping experience by suggesting related products. This leads to more repeat purchases and customer loyalty.

b) Loyalty programs:

Customers who are interested are more likely to join reward programs, which encourage them to buy again and spend more. Not only do these programs reward loyal customers, they also give brands more chances to connect with their customers, which makes them more interested (Qualtrics, 2022). *For example*, Starbucks' Rewards Program of how keeping customers interested leads to higher CLTV (Starbucks Reward, 2024). The program keeps track of what customers buy, what they like, and how often they visit, so it can offer them personalised deals and rewards based on how they act. This plan has worked very well so far: As of Q1 2022, Starbucks said that their Rewards Program brings in about 53% of the money that the business makes from its U.S. stores (PYMNTS, 2022). Customers who are involved in a loyalty program not only come back more often, but they also spend more each time, which improves CLTV.

c) Continuous engagement via omnichannel strategies:

Businesses can contact with customers on many channels, like mobile apps, social media, email, and in-store interactions. This makes it easier for customers to stay connected and interested. Customers have a more personalised experience the more touchpoints they deal with, which leads to long-term loyalty and higher CLTV (Kumar *et al.* 2023). *For example*, Nike's omnichannel approach shows how combining digital and real-world experiences can increase customer lifetime value (CLTV) and engagement (Pandey, 2024). The company keeps users interested with workout plans, fitness challenges, and personalised product ideas through its Nike+ app. Customers can use the app in shops to get special deals or make their purchases

more unique. Nike's strategy shows how long-term engagement across various channels can have a big effect on CLTV.

2.2.3 Models that integrate demographics with user behaviour data

In today's business world, one of the best ways to predict Customer Lifetime Value (CLTV) is to combine personal data with information about how users behave (Firmansyah *et al.* 2024). Businesses can make better and more complex models for figuring out how much a customer is worth and how to make their experience more unique by mixing these two types of data. Businesses can use this combination to find their most valuable customers and guess what they might do next, which helps them make the best decisions about their marketing budgets, strategies for keeping customers, and product suggestions.

Commonly used models for CLTV predictions are

a) RFM (Recency, Frequency, Monetary Model):

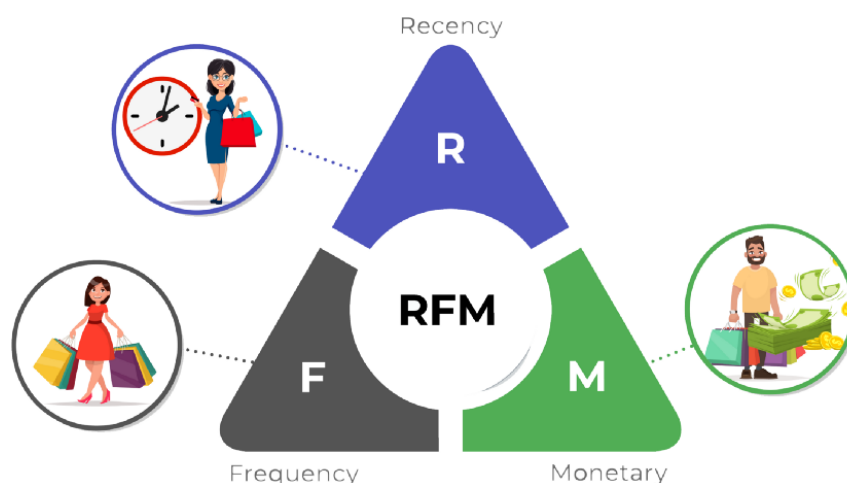


Figure 3 RFM (Recency, Frequency and Monetary)

(Source: Palati, 2021)

Standard CLTV calculations use the RFM model, which groups customers by when they last bought, how often they buy, and how much they spend (Ho *et al.* 2023). RFM is helpful, but it doesn't provide demographics or behaviour beyond purchases. Companies increasingly integrate social and behavioural data to RFM to improve accuracy. *For example*, Amazon's RFM research is better now that it includes both demographic and behavioural data, Amazon keeps track of more than just what customers buy (AWS, 2023).

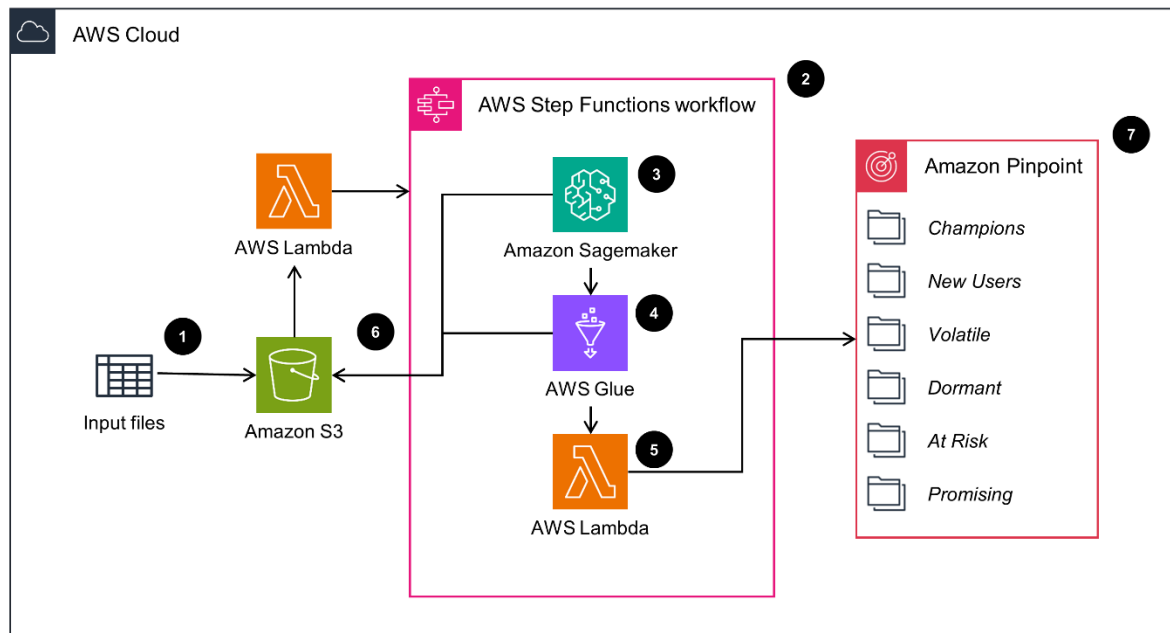


Figure 4 RFM analysis on Amazon Pinpoint

(Source: AWS, 2023)

They also keep track of how they browse, review products, and search. *For example*, a young worker in a city might get ads for business gadgets that are more relevant to them based on their demographics and how they act. Amazon uses this information to guess not only the current CLTV but also the customer's future product choices and how they will interact with the company. This creates a very personalised experience (AWS, 2023).

b) **Random Forest Models:**

A different machine learning technique, the Random Forest model, builds many decision trees and uses their results to forecast better. Big datasets can assess demographic and behavioural data well (Mohandoss *et al.* 2021).

Random Forest Classifier

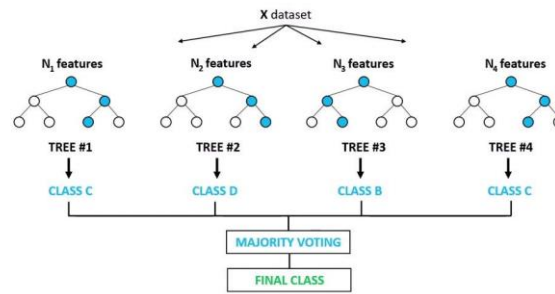


Figure 5 Random Forst Model

(Source: Chauhan, 2021)

Random Forest models can handle linkages and complexities that simpler models overlook since they look at several data kinds. Netflix employs Random Forest models to estimate a user's CLTV based on demographics like age, family size, and nation and behaviour data like viewing history, binge-watching, and content engagement (Khandelwal *et al.* 2023). Netflix uses these facts to recommend series and films that fit viewers' likes and habits.

c) **Gradient Boosting Models:**

Gradient Boosting Models (GBMs) are machine learning algorithms that create decision trees one after another, repairing faults in the previous trees (Aziz *et al.* 2020).

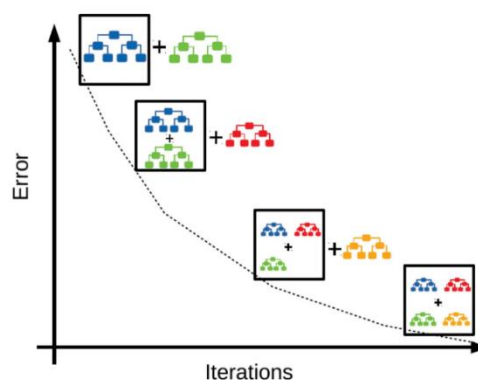


Figure 6 Gradient Boosting

(Source: Wei, 2024)

GBMs excel at large social-behavioural datasets, GBMs can identify more complex linkages between consumer attributes and future value by mixing structured demographic data with unstructured behaviour data like reviews and customer service interactions (Shyam *et al.* 2020). Uber employs GBMs to estimate rider and driver CLTV by combining demographic data like age, city, and job with behavioural data like how often clients use the service, when they use it, and when they stop (Uber Blog, 2022). It helps Uber determine which clients will become long-term customers and which need retention measures like tailored promotions or reward programmes. This methodology has helped Uber develop and maintain involvement.

2.2.4 Challenges in current CLTV estimation methodologies.

Customer Lifetime Value (CLV) is an important measure for businesses because it tells them how much money each customer will bring in over the course of their relationship with the business. Here are some of the biggest problems with the way CLTV is estimated now:

a) Inconsistent data quality and availability:

One of the biggest problems with getting a good estimate of CLTV is that there isn't enough consistent data (Faster Capital, 2023). CLTV models usually need a lot of different kinds of data, like a customer's buy history, engagement metrics, demographic data, and behavioural data. However, a lot of businesses either don't collect all of this data or have datasets that aren't full because of data silos. **Data Silos:** In many businesses, customer information is kept in different systems, like CRM, marketing automation, and e-commerce platforms, this makes it hard to get a perspective of the customer (Jarris, 2020). **Data Gaps:** it's hard to get a good estimate of CLTV when customer profiles are missing or not full, this is especially true for new customers who don't have many purchases or engagements in the past (Medium, 2023).

b) Change in customer behaviour:

Customer behaviour changes all the time, and CLTV models don't always show how customers' buying habits or tastes change over time (Czinkota *et al.* 2021). If the company try to figure out CLTV by looking at past data, it can be hard to do if customers' behaviour changes because of things like changes in the economy, actions by competitors, or changing customer wants. **Behavioural Shifts:** Customers can go from being very valuable to not very valuable or back again based on the situation, which is something that CLTV models often fail to predict (Faster

Capital, 2023). **Seasonality:** Demand changes with the seasons for many businesses, which makes it harder to figure out CLTV using models that don't take seasonality into account.

c) **Over-resilience on historical data:**

Most methods for estimating CLTV use a lot of data from the past to guess what will happen in the future. This method might not work well in fields where customer needs or market factors change quickly (Caldwell, 2022). If the company only look at facts from the past, the company might make wrong predictions because the company think that past actions are a good indicator of future actions. **Outdated Data:** In businesses that change quickly, market conditions and customer tastes can change quickly, making old data less useful (Chan, 2020). **Lack of Real-time Insights:** a lot of CLTV models don't use real-time customer data, like present engagement or social media interactions, which can give a more accurate perspective of how likely a customer is to keep buying.

2.2.5 *Solutions enhancing accuracy in CLTV forecasting models.*

a) **Inconsistent data quality and availability:**

Companies need to spend money on building a unified customer data system and making sure data integrity across all channels to fix the problem of inconsistent data quality and availability. **Set up unified data platforms:** Businesses can make a single source of truth for customer data by combining Customer Relationship Management (CRM) systems, marketing automation tools, and e-commerce platforms (Salesforce, 2022). Customer Data Platforms (CDPs) help put together info about a customer from all of their interactions with a business. **Data hygiene practices:** businesses should check and clean their data on a regular basis to make sure it is correct, full, and consistent. Automated data cleaning tools can get rid of unnecessary records, add missing data, and make sure that all datasets use the same style (Cdata, 2024).

b) **Change in customer behaviour:**

Businesses should use dynamic CLTV models that can change expectations in real time based on how customers are interacting with them. This way, businesses can account for changes in how customers act. **Use dynamic CLTV models instead of static ones:** Dynamic CLTV models can adapt to changes in how customers behave by constantly updating expectations based on what customers have done recently (Pandey, 2024). Machine learning systems can look at data

in real time to make estimates more accurate as new data comes in. ***Use behavioural analytics:*** Companies can see changes in their customers' interest and involvement by keeping an eye on key behavioural metrics like website visits, email engagement, or app usage. These actions that don't lead to a purchase reveal important information about future buying plans.

c) Over-resilience on historical data:

It's not enough to just use old data, since it doesn't always show how the market and users have changed. In order to deal with this, businesses should think more ahead of time. When businesses add real-time data about how customers connect with them, they can quickly adapt to changes in market conditions and customer behaviour. This is because cloud-based and advanced analytics tools can handle data in real time, giving the company the most up-to-date information. Planning for different possible futures and running models to guess what will happen in the market can help businesses figure out how customers might behave in those futures. This helps the business handle quick changes in the market, like when the economy crashes or a new competitor releases a product.

2.3 Literature Gap

New Customer Lifetime Value (CLTV) research can help businesses use behavioural analytics and demographic data to predict customer profitability. However, certain gaps remain, one issue is a lack of studies on customer behaviour over time. Many CLTV models use old data and ignore how customers' preferences and brand interactions evolve over time. This makes it impossible to predict value in fast-changing industries like tech and e-commerce.

Additionally, most models simply include purchase behaviours, they ignore the importance of social media, ideas, and business promotion. Despite their significant impact on customer value, most CLTV assessments exclude these acts. The worst part is that few high-tech AI and machine learning technologies can quickly increase CLTV estimates. Some study has been done, but not enough on how real companies use these technologies.

Finally, multichannel identification issues are understudied, when firms struggle to track and value customers across online and offline touchpoints, these occur. This complicates CLTV calculations in omnichannel environments.

2.4 Conceptual Framework

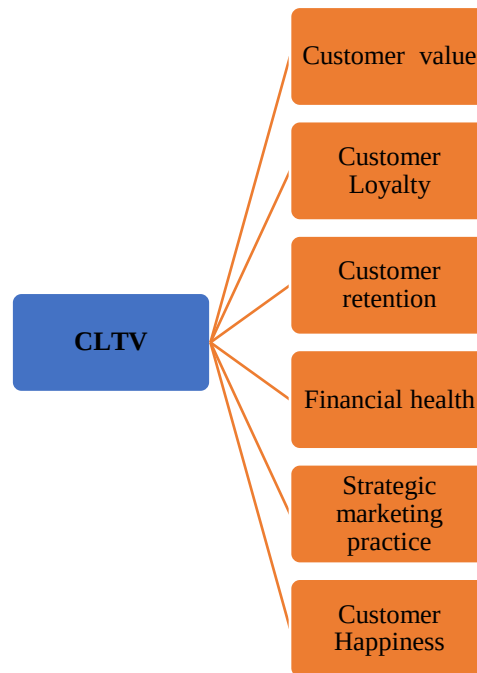


Figure 7: Conceptual Framework

(Source: Self-Developed)

2.5 Contribution

As a result of combining demographic and usage data with machine learning models, this study adds to the knowledge of how to estimate Customer Lifetime Value (CLV). It talks about how personalised customer interaction strategies can change CLV predictions and suggests better methods to make forecasting more accurate. The study also points out problems with the way CLV is estimated now and suggests ways to fix them, like dealing with bad data, changing consumer behaviour, and relying too much on old data. This gives companies useful information they can use to make their marketing and customer retention strategies more accurate and data-driven.

2.6 Limitations

One problem with this study is that it depends on data quality and availability, which can be very different between organisations. If datasets are inconsistent or missing parts, it could make CLV forecasts less accurate. Another problem is that machine learning models may not be able to fully account for sudden changes in market trends, customer behaviour, or economic factors,

even if they are very smart. As the models may be based on a lot of old data, the study's results may not be as useful in fields where customer tastes change quickly.

2.7 Why it is important

This study is important because having a good idea of the CLV helps companies use their resources wisely, focus on the most valuable customers, and make marketing plans that help them keep those clients. The study shows that companies can better guess what their customers will do in the future by using machine learning and social and behavioural data together. Enhanced CLV forecasting models give businesses information about how engaged their customers are, which helps them make more money through personalised interactions, dynamic retention strategies, and smart marketing investments, all of which are very important in today's data-driven economy.

2.8 Critical analysis

This study does a good job of looking at how to combine machine learning and CLV prediction, but one big problem is how to make models that are changeable and can change all the time to fit changing customer behaviour. In addition to this, depending on prediction analytics and real-time data could require a lot of resources, which could make it hard for smaller businesses to use. Therefore, machine learning models can give you useful information, but they need a lot of data processing power, and they can only work if you have a lot of high-quality data from all of your interactions with customers.

2.9 Risk factors

Some risks that come with estimating CLV include relying too much on past data, which might not always properly show how customers will behave in the future. Poor data quality or sources of fragmented data can also make machine learning models less accurate, which could lead to wrong conclusions. Long-term CLV predictions could also be wrong if market conditions change without warning, the economy slows down, or competitors take actions. Overfitting models to past behaviour without taking into account how unclear the future may be may also make it more likely that marketing resources will be wasted.

2.10 Chapter Summary

This chapter talks about how to figure out the Customer Lifetime Value (CLTV) by using machine learning to combine demographic data with usage trends. Customer involvement is strongly correlated with CLTV, as shown by Amazon, Starbucks, and Netflix. In this chapter, advanced models employ demographics and user behaviour to forecast CLTV more accurately. These models account for complex customer behaviour. Current CLTV calculation concerns include inconsistent quality data, shifting client behaviour, and attribution challenges in numerous situations. These issues can be addressed using unified data systems and dynamic CLTV models. Finally, it highlights research gaps, including the need for models that capture real-time customer behaviour and account for non-CLTV actions.

Chapter 3: Research Methodology

3.1 CRISP DM

The CRISPDM the Cross-Industry Standard Process for Data Mining (CRISPDM) is a framework on how to build machine learning models (Saltz, 2021). It includes 6 steps: Business Understanding, Data understanding, Data Preparation and Model evaluation & Deployment. The process is repeated, allowing data scientists to improve and enhance their models through repeated pattern. CRISP-DM is widely used because of its ability and flexibility to make different types of data mining and analytics project.

The CRISP-DM approach is divided as follows:

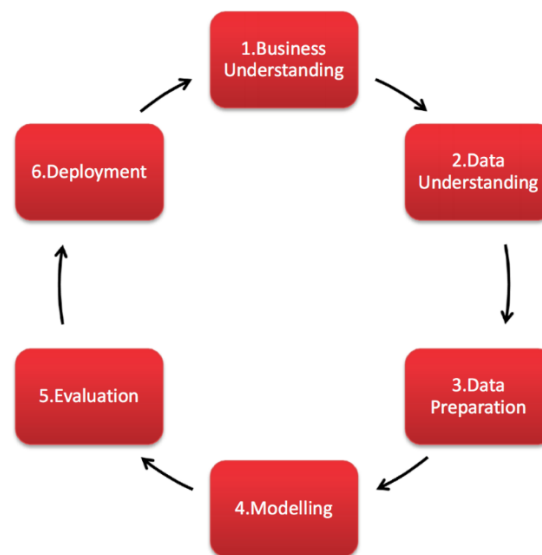


Figure 8 CRISP-DM

(Source: Self-developed)

3.1.1 Business Understanding

This step focuses on understanding the project goals, objectives, and requirements from a business viewpoint, then converting this information into a data mining problem definition and an early plan with a specific set of tasks and want result to achieve project-level target.

3.1.2 Data Understanding

The dataset used in this work begin from an insurance company and is formed to check customer behaviour and policy-related features. It includes detailed records of 98,415 policy owner, covering different parts such as financial information, demographics claims, and

customer lifetime value (CLTV). The dataset contains 12 variables. These variables cover unique identifiers for each policy owner, also with their gender, geographical area, educational qualification, and income level (Kaggle, 2024). The marital status of the policy owner is defined as integers, although the number of years since the policy owner joined (vintage) is provided as a numeric value.

Financial parts are captured through the total claim amount made by each policy owner, and the number of policies they hold is defined as a categorical variable. These express types of insurance policies, as well as the large policy categories, are also provided. Customer lifetime value (CLTV) is one of the leading metrics which defined the rated total value that a customer brings to the company over their lifetime.

This dataset is specifically used for analysing customer behaviour, claim patterns, and policy performance. It allows for customer division, churn prediction, and the identification of key factors have an effect on CLTV. Also, it helps in understanding the relationships between policy owner demographics and their claim behaviour, making it valuable for risk management and customer holding strategies.

EDA

1. Count of Gender

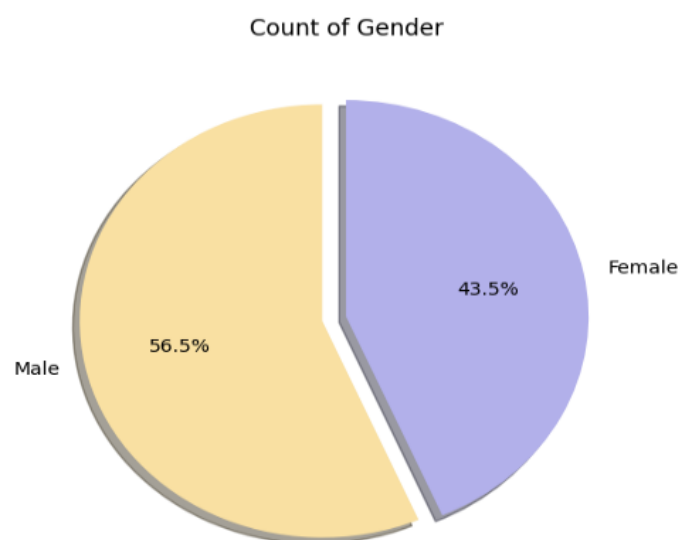


Figure 9 Count of Gender

(Source: Self-developed)

The Pie chart in above Figure shows the distribution of gender between clients. the chart shows the small imbalance, with 56.5% of the clients are male and 43.5% are female. This difference might be important to keep in mind when analysing trends by gender or developing gender-specific models to keep results fair.

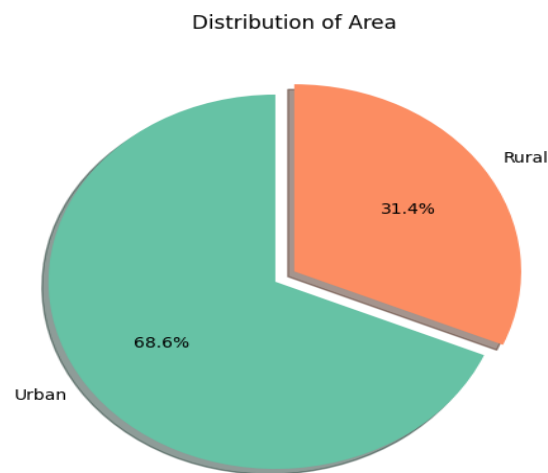


Figure 10 Distribution of Area

(Source: Self-developed)

The pie chart in figure 2 shows the distribution of Area. The pie chart represents the distribution between rural and urban areas. This distribution shows that the 68.6% of the population are from urban regions and with 31.45 are come from rural areas. This big difference identify that the majority of the population settle in urban areas. when analysing the data, it's important to keep this difference in mind, as it could affect trends in urban and rural growth, access to services, and economic differences in each area.

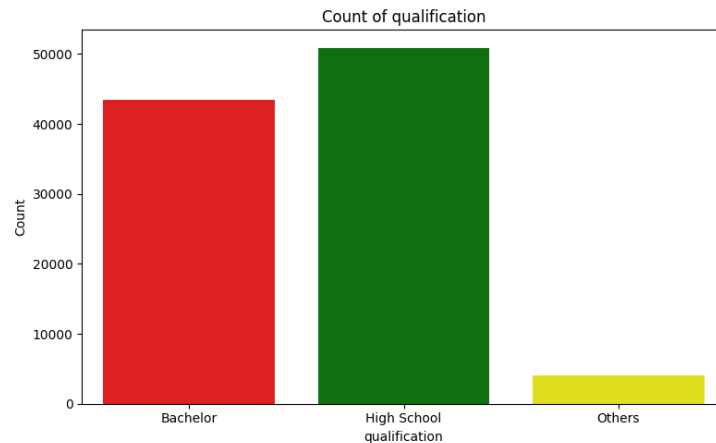


Figure 11 Count of qualification

(Source: Self-developed)

The bar chart in above Figure 3 shows the different qualifications levels in the population. The largest group is high school qualification and the second largest group is bachelor’s degree, and the smallest group is “other” qualifications. This shows that the most of the people have either “high school” or “bachelor’s” education, only few people choose the “other” type education. This difference can impact job opportunities , earning potential and decisions about education policies .

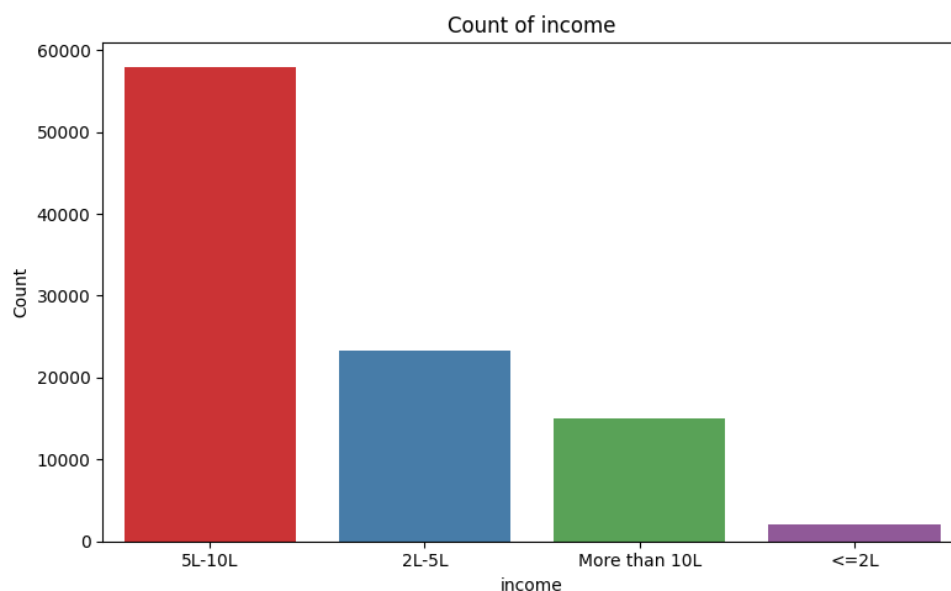


Figure 12 Count of Income

(Source: Self-developed)

The bar chart in above figure 4 shows the distribution of income levels. Most of the people earn between 5L-10L income, while few people earn between 2L-5L or more than 10L. The smallest group earn 2L or less. The trend towards higher incomes 5L-10L, shows the income growth of the population, which can affect purchasing power, financial habits, and markets targeting strategies. Understanding that this income distribution can help in creating products and services suitable for different income levels.

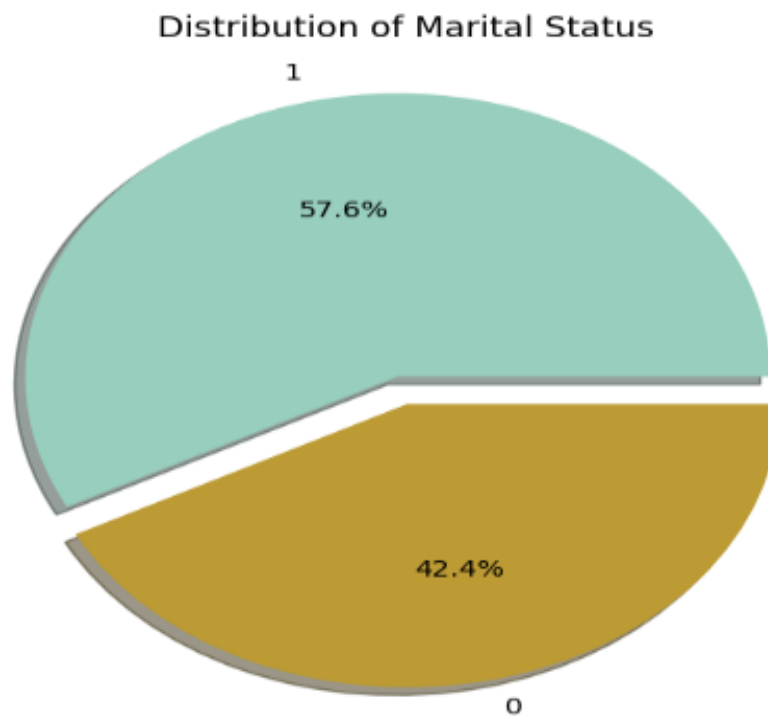


Figure 13 Distribution of Marital Status

(Source: Self-developed)

The pie chart in above figure 6 shows the distribution of marital status among the population. The distribution shows that the largest portion of the population is marital status with 57.1% (coded as 1), and the unmarried status with 42.4% (coded as 0). This distribution can help us to understand the things like household spending and lifestyle choices as being married can affect the financial decisions and product preferences. Business can use this information for better match to their products or services to different groups.

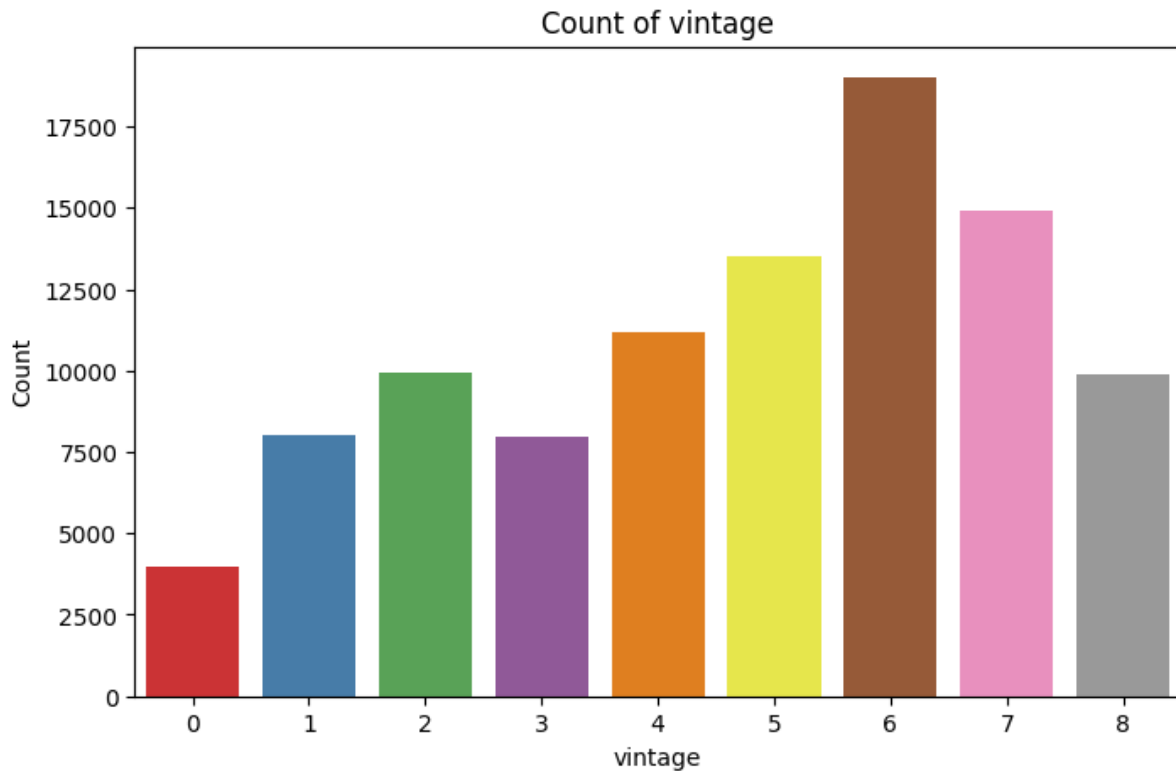


Figure 14 Distribution of Vintage Count

(Source: Self-developed)

The bar chart in above figure 7 shows the count of individuals across different "vintage" categories, which represent a measure of tenure or duration. The highest number of individuals, approximately 17,500, belong to the vintage category 6, followed by category 7 and 5. Lower vintage levels (0-2) have fewer individuals, indicating that fewer people have shorter tenures. This skew towards higher vintage categories suggests that a significant portion of the population has longer-term experience or engagement. Such data is valuable in understanding customer loyalty, financial product tenure, or retention patterns, helping organizations design targeted strategies for customer engagement and retention.

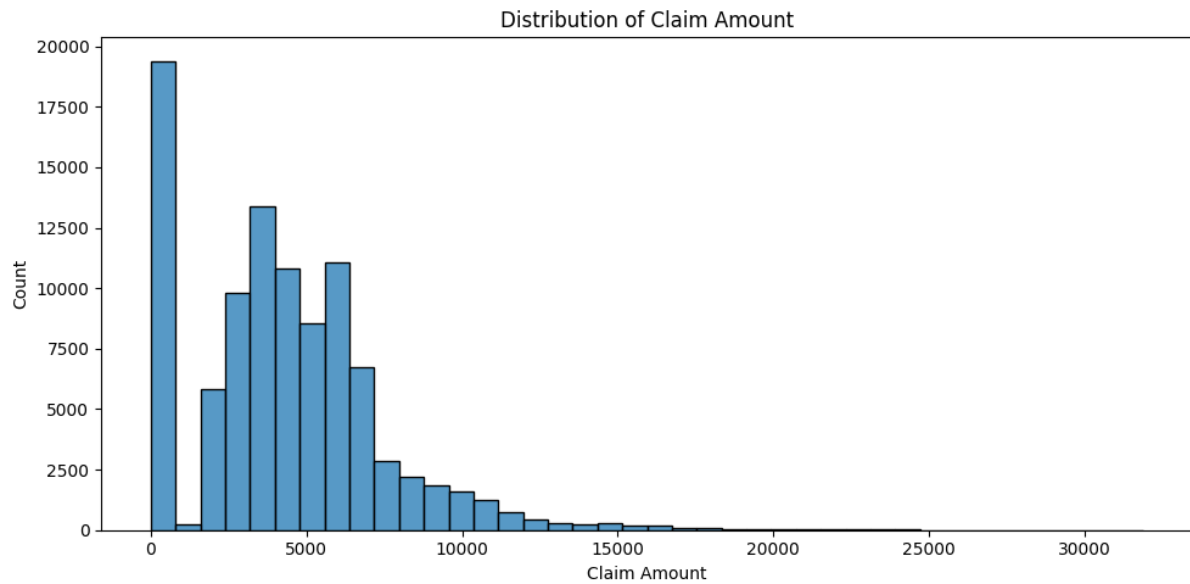


Figure 15 Distribution of claim Amount

(Source: Self-developed)

The bar chart in above figure 8 represents the distribution of claim amounts across a dataset. The highest frequency of claims is seen at the lower end of the spectrum, with most claims being small in value, clustered around 0 to 2,000. As the claim amounts increase, the frequency drops, showing fewer claims above 10,000. The tail at the right suggests that while there are higher claims (up to 30,000), they are much less common.

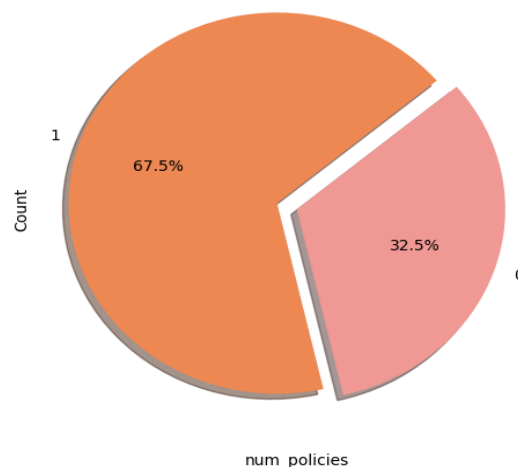


Figure 16 Distribution of Num policies

(Source: Self-developed)

This Pie chart in above figure 9 shows the distribution suggests that a majority of individuals are engaged with the insurance system, having at least one policy. However, the substantial minority without any policies represents a potential target market for insurance providers. Understanding this segmentation can help insurers in creating strategies to increase policy adoption, particularly among the 32.5% who have yet to invest in any insurance products.

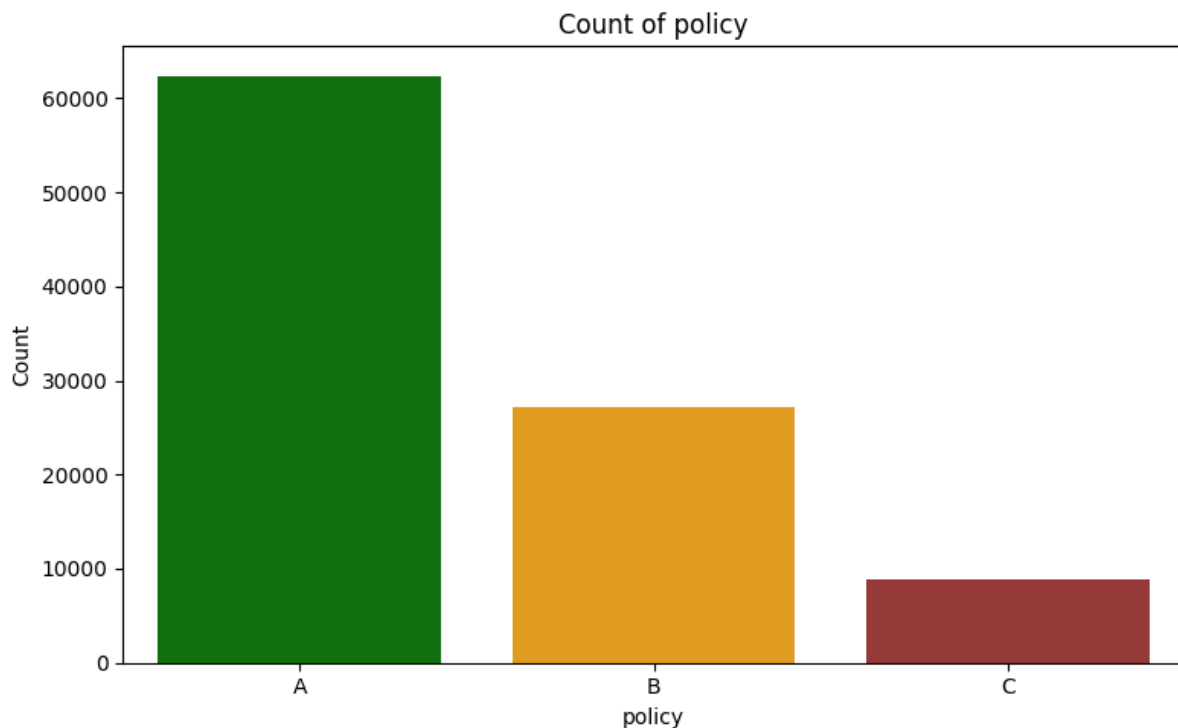


Figure 17 Distribution of different policy types

(Source: Self-developed)

The bar chart in Figure 10 shows the distribution of different policy types (A, B and C) based on the number of policies. Policy A is the most popular policy, with 60,000 individuals holding this policy because of its benefits, affordability and accessibility. Policy B, comes in second, with 25,000 holders while Policy C has the least engagement with 10,000 holders.

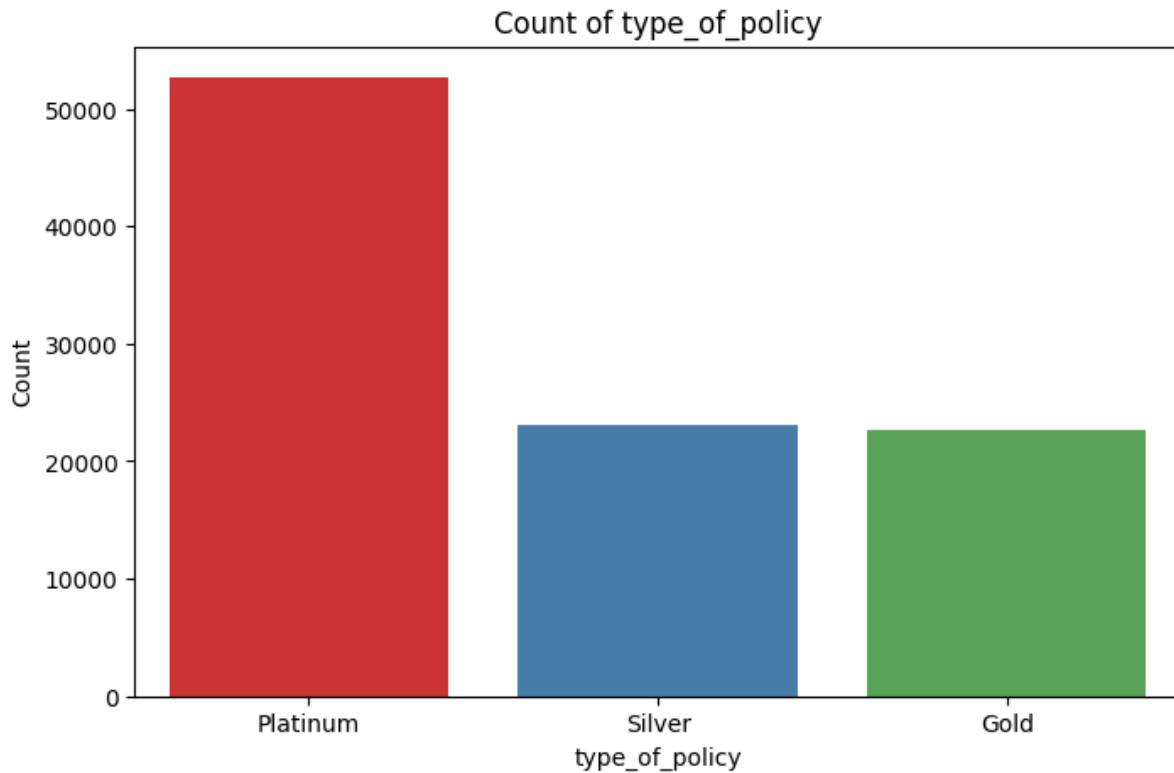


Figure 18 Distribution of type of policy

(Source: Self-developed)

The bar chart in above figure 11 shows the distribution types among customers. The most common policy is 'Platinum' policy with over 50,000 customers, while the 'Silver' and 'Gold' policies have visibly fewer customers, each about half of the count of 'Platinum'. This indicates that there is a high demand of 'Platinum' policy because of its benefits and features. After viewing, this distribution can help the business to focus on the popular policy types and explore the ways to make 'Silver' and 'Gold' policy more attractive to consumers.

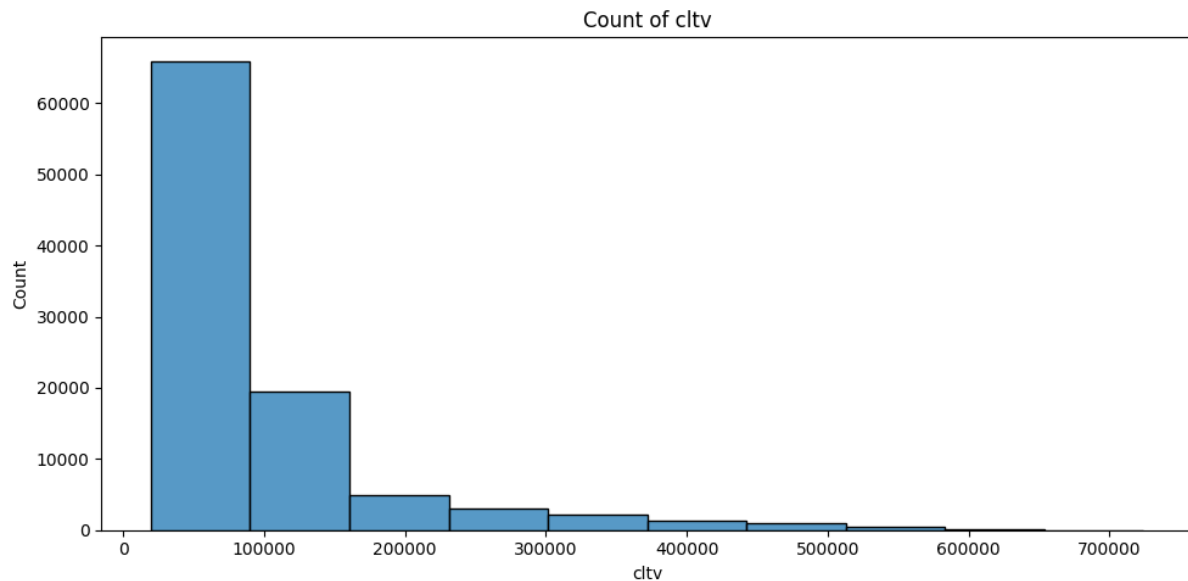


Figure 19 Distribution of CLTV

(Source: Self-developed)

The Histogram in above figure shows the distribution of Customer Lifetime Value (CLTV) among clients. Most of the customers have CLTV lower than 100,000, with the number of customers decreasing speedily as the values of CLTV is increases. Only few customers have CLTVs over 200,000, and very few reach the higher end of the range, close to 700,000. This skewed distribution shows that the most of the business's revenue opportunities are contained in customers with lower CLTV, this highlights a potential area for targeting higher-value customers or improvement in customer value growth strategies.

3.1.3 Data Preparation

The Data Preparation is the process of raw data cleaning and transforming It as needed for analysis or modelling (Brownlee, 2020). Handling missing values, scaling, encoding your data in a way that improves model performance and accuracy are all part of it.

1. Log Transformation: The Log Transformation is a mathematical process whereby you apply the logarithm to each point of data to reduce its skewness (Lee, 2020). It normalizes highly skewed distribution, in particular variables which increase exponentially or vary in magnitude.

2. Mapping: Mapping is simply replacing categorical values to numerical equivalents, helping machine learning algorithms parse more easily (Caplan *et al.*, 2016). Often it is used for data with natural order, like low, medium, high, 1, 2, 3.

3.Label Encoding: Label Encoding creates such conversion from categorical values to numerical labels: e.g., from "Red," "Green," and "Blue" to 0, 1, and 2, respectively (Jia and Zhang, 2021). This is especially important for numerical input methods such as tree-based methods.

They aid in cleaning up this data so your machine learning models can function better.

3.1.4 Modelling

During the Modelling phase, a Decision Tree Regression, an AdaBoost Regression, and a linear regression are developed. These algorithms are learned and tested to estimate outcomes or recognize patterns and optimize performance based on the characteristics of the inputted data and the intended goals.

Random Forest

A *random forest* is a supervised algorithm that uses an ensemble learning method. It is a collection of multiple decision trees, the output of which is the consensus of the best answer to the problem. Random Forest can be used for classification or regression (Daghistani and Alshammari, 2020). Random Forest follows specific rules for tree growing, tree combination, self-testing and post-processing. It is robust to overfitting and is considered more stable in the presence of outliers and very high dimensional parameter spaces than other machine learning algorithms.

$$\text{Majority Vote}(\text{Tree1}(x), \text{Tree2}(x), \dots, \text{Tree } n(x))$$

AdaBoost Regressor

The AdaBoost Regressor is a real-time booster algorithm that is used to generate an ensemble model of multiple weak learners, where the weak learners are mostly decision trees (Hussain, and Zaidi, 2024). Spectral clustering works in an up-temporal manner; the current one addresses the errors the previous models committed. In the context of regression, AdaBoost computes values that seek to minimize the loss function by paying a relatively large penalty to data points with significant errors so as to prioritize them in the subsequent iterations.

This algorithm minimizes both the bias and variance, hence improving the model's ability to detect features in the data set. AdaBoost Regressor is suitable for use with datasets with different features of significance and does well in conditions where outright single models are

not sufficient (Haque, *et al.*, 2022). However, it is noisy and unstable because, in the case of extreme values or outliers, they are assigned high levels of rates or weights. On the other hand, parameter optimization and the format of data to be trained on are essential and should be done delicately.

Linear Regression

Linear Regression is one of the most basic statistical and machine learning models applied in predicting an interval or ratio level dependent variable in terms of one or more predictor variables (Maulud, and Abdulazeez, 2020). It exemplifies a straight-line relationship between the dependent variable, in this case, output and the independent variables, or inputs and plots them as the regression line.

Linear Regression is a very interpretative, numerically simple model that is optimal for gaining an understanding of correlations or tendencies as well as making simple forecasts. It constrains linearity, independence, homoscedasticity (the variance of errors is fixed), and residual normality (Valdés-Souto, and Naranjo-Albarrán, 2021). However, it can suffer from multicollinearity problems, the presence of outliers and nonlinear relationships at times. Despite that, Linear Regression still remains one of the most important techniques for predictive modelling and serves as a reference point for more complex algorithms.

Decision Tree regressor

Decision Tree Regressor: It is a machine learning algorithm for continuous outcome variable. It divides a dataset to create a tree, with each node being a feature and each branch a decision rule based on the values of the feature. Predictions at the leaf nodes is the mean of the target variable in that subset, and splits minimize the spread using metrics like mean squared error (MSE). It is very interpretable, performs well with both numerical and categorical data, and can capture non-linear relations (Kushwah, *et al.*, 2022). But too much depth leads to over-fitting and is countered by means of post-pruning or boosting in a more robust manner.

XGBoost Regressor

XGBoost Regressor: A super-fast and generalizable, scalable boosting based machine learning technique. It builds an ensemble of decision trees in a stage-wise manner, where each tree improves upon the errors of the previous trees. It overcomes over-fitting and

improves generalization using techniques like regularization, shrinkage, and column subsampling. XGBoost deals with missing values, works on both numerical and categorical data, and masterfully captures complex non-linear relationships. And clearly known for the fact that it can scale out to work with huge datasets very efficiently computationally. Due to its flexibility and competitive performance, it has become a web industry standard for regression problems of all types and at large scale.

3.1.5 Evaluation

In the Evaluation phase, model performance is evaluated along with other performances, such as accuracy, the R-squared factor, which describes the fraction of the prevent variance for the model, and RMSE and MSE, which measure prediction accuracy or errors.

R-squared

R-squared is a statistical measure used to describe how well data fit a mathematical model, i.e., regression model, and can range from (least square) 0 to 1, where a higher number indicates a better fit (Onyutha, 2020). It varies between 0 and 1, where the value closer to 0 shows that the applied model does not explain any of the variance, and the value closer to 1 suggests the model describes the entire variance. A closer look at the model reveals that the value of R-squared indicates that the model fits closely to the data. Still, it means that R-squared should be used with great caution, especially if the model is very complicated or if there are many predictors, yet the model is actually not all that good.

RMSE

Root Mean Squared Error (RMSE) is one common measure of model accuracy for a regression type of equation (Chicco, Warrens, and Jurman, 2021). It estimates the error standard by using the square root of the mean of squared differences of actual from the predicted. RMSE offers a way of quantifying the deviations from the exact values, and the higher the RMSE, the worse the model. The strength of RMSE is that significant errors are down-weighted due to the squaring of differences, hence making the technique very cautious of outliers. It is measured in the same form as the target variable, and therefore, it can be easily translated into to real-life values and contexts.

MSE

Mean squared error, as is known, is also one of the types of assessment of the accuracy of the regression model (Diedolo, *et al.*, 2022). It is known that it is computed by taking the mean of the square differences between the predicted values and the actual ones. MSE gives the average extent of deviation of the model's predictions from its actual value, which denotes poor performance based on more significant values. MSE is reflective of outliers because squaring of the differences increases with significant errors. It also evidences that a lower MSE suggests higher achievable predictive accuracy. MSE is expressed in squared units of the target variable, which is not convenient but displays some differences between RMSE and MSE.

3.1.6 Deployment

As part of the deployment phase, several models, namely Decision Tree Regressor, AdaBoost Regressor, and Linear Regression, are compared. As this work entails comparing models, no deployment will be made to any platform here.

3.2 Tools used

The tools used for the current project are Jupiter Notebook, Python language as well, and a particular system with accurate ingredients called configuration. Jupiter Notebook is a web-based interactive computing tool that enables developers, scientists and data analysts to write and execute code, interactive definition, and visualization of data as well as enhanced formatted textual descriptions in the same document in a faster manner and any programming language (Johnson, 2020). Because it is more flexible and easier to use, it is frequently used for large-scale data science and machine learning.

The main software development language is Python, with version 3.9.7, which is used to develop machine learning models and data analysis (Bourne, *et al.*, 2023). To make different computations required in the process of constructing models, Python has available multiple libraries such as NumPy, Pandas, and scikit-learn. The hardware used includes a computer with Windows 10 Pro as its operating system and 16GB RAM, which allows complex computations in datasets and model training.

3.3 Chapter Summary

This chapter utilizes machine learning algorithms to analyse a dataset of customers from an insurer and make CLTV predictions. The data includes customer demographics, financial records, and CLTV. The CRISP-DM methodology guides the process, encompassing steps such as business understanding, data preparation, and model deployment.

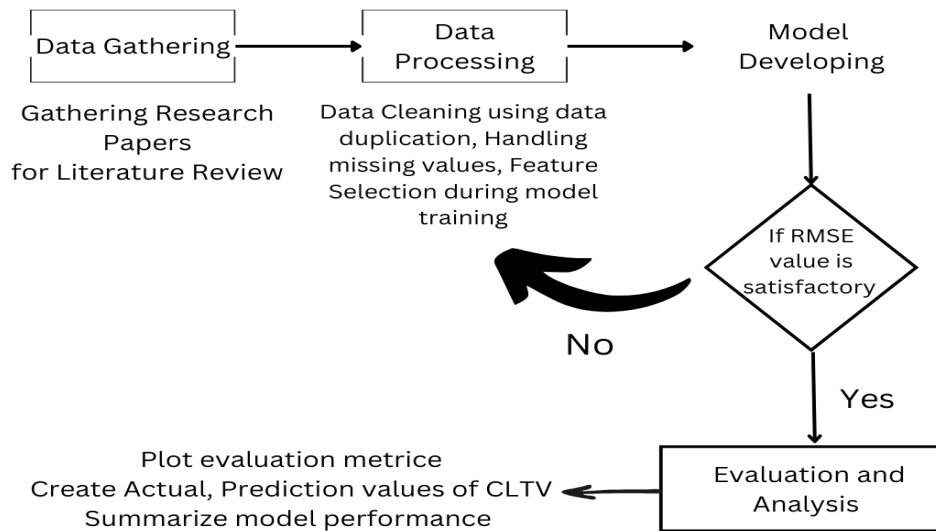


Figure 20: Customer Lifetime Value (CLTV) Model Development Process

(Source: Self-developed)

The accompanying flowchart, 'Customer Lifetime Value CLTV Prediction Workflow,' visually represents these steps, detailing data gathering, processing, model development, and evaluation. Three models—Decision Tree Regressor, AdaBoost Regressor, and Linear Regression—are developed and their performance evaluated using metrics like R-squared, Root Mean Squared Error (RMSE) and Mean Squared Error (MSE). This comparison identifies the most accurate model for predicting customer behaviour

to support business decisions like risk control and customer loyalty initiatives. Tools such as Jupyter Notebook, Python, and a Windows 10 Pro system equipped with powerful hardware ensure efficient handling of big datasets and model training.

Chapter 4: Result and Analysis

The Result and Analysis section contains the results of the study or experiment, as well as their interpretation. Key data is summarized using tables, charts or graphs to present results clearly and concisely. Here, it points out trends, patterns and comparisons from the data. It interprets the results in the analysis part. It highlights significant results, relations, or differences, interprets how these results fit the goals, hypotheses, or previous work and considers their significance. This section gives a description, no subjective opinions, and an interpretation from which it can logically conclude with some generated insights from the provided dataset.

4.1 Linear Regression

Model	RMSE	R-Squared	MAE	MSE
Linear Regression	33732.76	0.85	23619.33	1137899557.54

Table 1 Performance Report of Linear Regression

(Source: Self-developed)

The Linear Regression model shows the RMSE of 33,732.76 that define the average margin of the predicted CLTV with the actual values. The model also provides the R-squared value of 0.85 which means the model is adequately fit to perform 85% to predict the Customer Lifetime Value (CLTV) variance. Nevertheless, the value of RMSE, being slightly higher than in case of males, indicates that there is potential in improving the predictive functionality. The MAE value of 23,619.33 also supports the mean difference between the real and the prognosis values. The estimated total sum of squares, MSE of 1,137,899,557.54, shows the size of errors in the same scale, and possibly non-linear relations not adjusted by the linear model above. Nevertheless, the model still proves rather successful for the identification of CLTV and can be used as an initial approach.

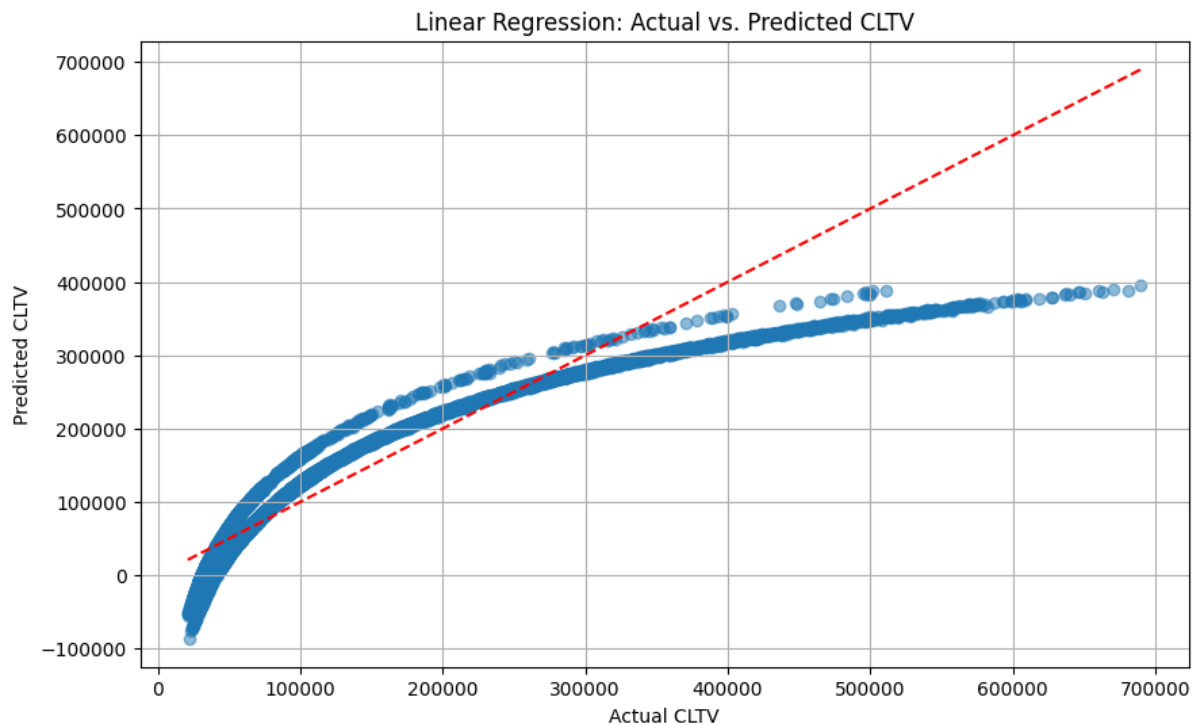


Figure 21 Linear Regression: Actual vs. Predicted CLTV

(Source: Self-developed)

The Above graph displays Linear Regression model results in the form of a scatter plot of the actual versus predicted CLTV. Actual CLTV values are plotted on horizontal axis while Vertical axis represents the predicted CLTV values. Blue dot indicates an individual data point. In an ideal world, all the points should lie very close to the red dashed line that is the actual - predicted line.

From the plot, it can be deduced that the model is relatively accurate for lower value of CLTV but prediction error increases as the actual CLTV value increases. This may mean there are problems with the assumed model, underfitting as the CLTV increases or restriction of the instantaneous linearity of the regression equation. A deviation from the perfectly straight learning curve signifies that more academic attention should be paid to feature extraction, nonlinear relationships and outlying cases.

4.2 Random Forest

Model	RMSE	R-Squared	MAE	MSE
Random Forest	50.10	0.99	5.31	2510.77

Table 2 Performance Report of Random Forest

(Source: Self-developed)

The Random Forest model is quite accurate based on the RMSE of 50.10, proving that on average the model is slightly off when predicting the Customer Lifetime Value (CLTV). From this it can deduce that the model is capable of predicting fairly closely to the real values with little margin of error. The R-squared of (0.99) indicated great fit by translating to the extent of 99% of CLTV variation within the data. Furthermore, a MAE of 5.31. This means that the average error of the model is 5.31, that is, the model is accurate as it is off by 5.31 unit, making it more reliable. The test result shows that, the MSE of 2,510.77 of the models implies that the sum of the total errors is relatively low to indicate that the Random Forest model is good for CLTV forecasting with high accuracy. As demonstrated in this performance, this model is a good candidate for analysing customer data in business analytics because it handles nonlinear patterns so well.

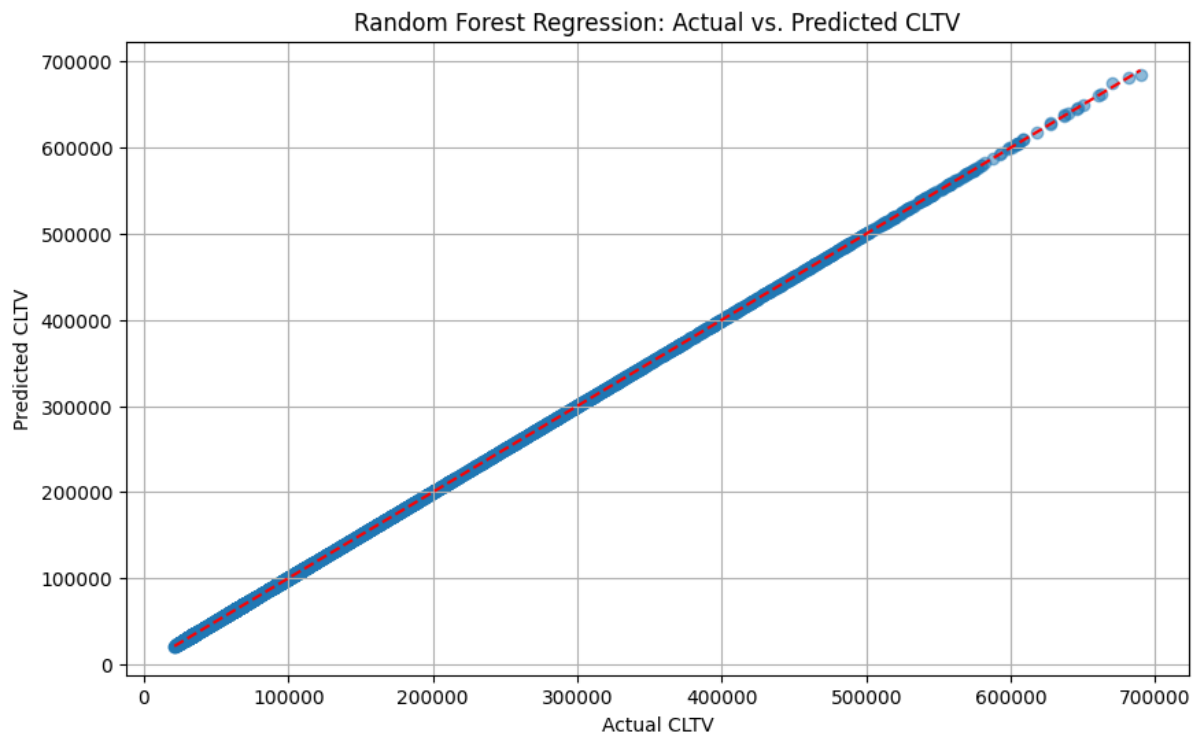


Figure 22 Random Forest Regression: Actual vs. Predicted CLTV

(Source: Self-developed)

Random Forest regression model Actual vs. Predicted CLTV (Customer Lifetime Value) is depicted in the chart above. These jumps represent remarkable performance of the Random Forest model where the predicted points gathered at the density of blue dots are located near the red dashed line that demonstrates ideal prediction.

Using the Random Forest, the accuracy of the model can be considered high since the gap between the actual and CLTV prediction values is rather small. This alignment even more enhances the model given it achieved high R-squared of 0.99 and a low RMSE of 50.10 to clearly mean it can identify and reflect various patterns and trends in the data appropriately hence enhancing flexibility. It was observed that Random Forest has acceptable accuracy for predicting CLTV due to this performance in practical use.

4.3 Decision Tree Regressor

Model	RMSE	R-Squared	MAE	MSE
Decision Tree Regressor	306.40	0.99	95.52	93886.27

Table 3 Performance Report of Decision Tree Regressor

(Source: Self-developed)

The performance of the Decision Tree Regressor model is quite good as the RMSE turns out to be 306.40 meaning moderately high average error in Customer Lifetime Value (CLTV) prediction is acceptable. Although there was this mild hitch, the R-squared of 0.99 implies that much variance in target variable is explained by the model, thereby implying a good the model has in explaining variance in future values. This is further supported by the MAE of 95.52, suggesting that the model is typically wrong by approximately 95.52 units. In addition to this, 93,886.27 it get the Mean Squared Errors (MSE) that gives an overview of the overall error of the model. Despite the capability of achieving the maximum accuracy, the Decision Tree model requires improvement in MAE, which in comparison to the other models is relatively higher; such improvement can be done through running or tuning of the hyperparameters. However, given its great R-squared and appropriateness in identifying connections essential in data, it is appropriate to use this sort of model for estimating CLTV.

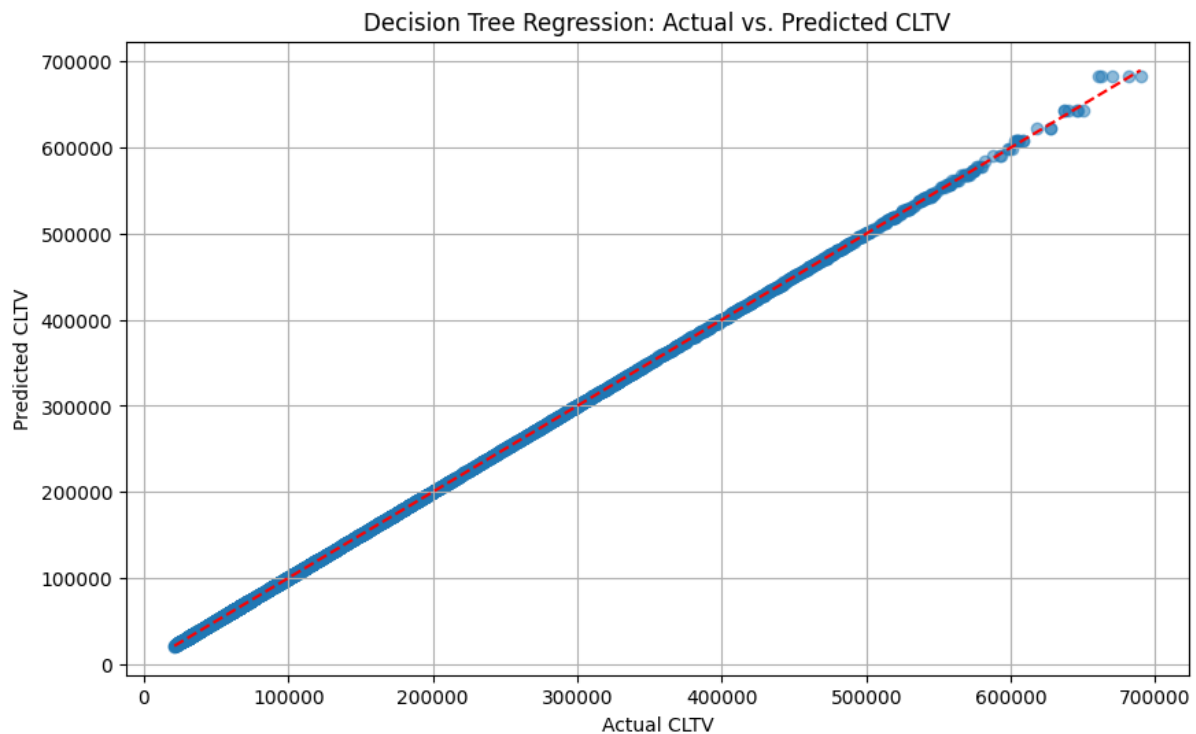


Figure 23 Decision Tree Regression: Actual vs. Predicted CLTV

(Source: Self-developed)

The graph Above shows the performance of the Decision Tree Regression model for the estimation of CLTV. The first graph is the scatter plot, mapping one actual CLTV value with one predicted CLTV value, shown on the y-axis, with x-axis representing the actual CLTV value; The red diagonal line represents the case if prediction was perfect.

Hence the plot indicates that almost all the data points are close to the diagonal thus pointing toward high accurate model. This is further evident from the fact that the model has a high Adjusted R-squared value of 0.99 meaning that the model method explains nearly all change in the datasets and a low RMSE of 306.40 indicating right approximations.

In this case, the use of Decision Tree Regression is advisable because it works well with non-linear characteristics with the help of splitting the data space into regions and thus making it easier to find explanations. From this model, it can be stated that how well it performs when operating on well-structured data such as those used in the CLTV prediction. Although the assessment on the validation set shows the promising results, external validation is strongly encouraged to avoid overfitting.

4.4 AdaBoost Regressor

Model	RMSE	R-Squared	MAE	MSE
AdaBoost Regressor	12117.86	0.98	8116.48	146842709.33

Table 4 Performance Report of AdaBoost Regressor

(Source: Self-developed)

The AdaBoost Regressor has an RMSE of 12,117.86 which means that the model was, on average 12,118 units off from the correct responses. The fit statistics also show outstanding details, with an R-squared value of 0.98 implying that 98% of the variability in the target variable, CLTV (Customer Lifetime Value) is attributable to the design of the model, which also confirms high predictive accuracy of the model. The obtained MAE: 8,116.48, which means that on average, the model over or under-predicts outcomes by 8,116 units. The values indicate that the MSE of 146,842,709.33 offers the total quantity of error in all predictions. On the aggregate, AdaBoost is much more accurate and a fairly good model by most standpoint, even if reflected by an RMSE that is marginally higher by the standards of the given dataset and other boosting models hints at the potential for further fine tuning of the parameters or for experimenting with other forms of boosting. Still, the accepted benchmark for R-squared is up to 0.98 while the MAE of the model is 1109.38 which makes it rather accurate in predicting CLTV.

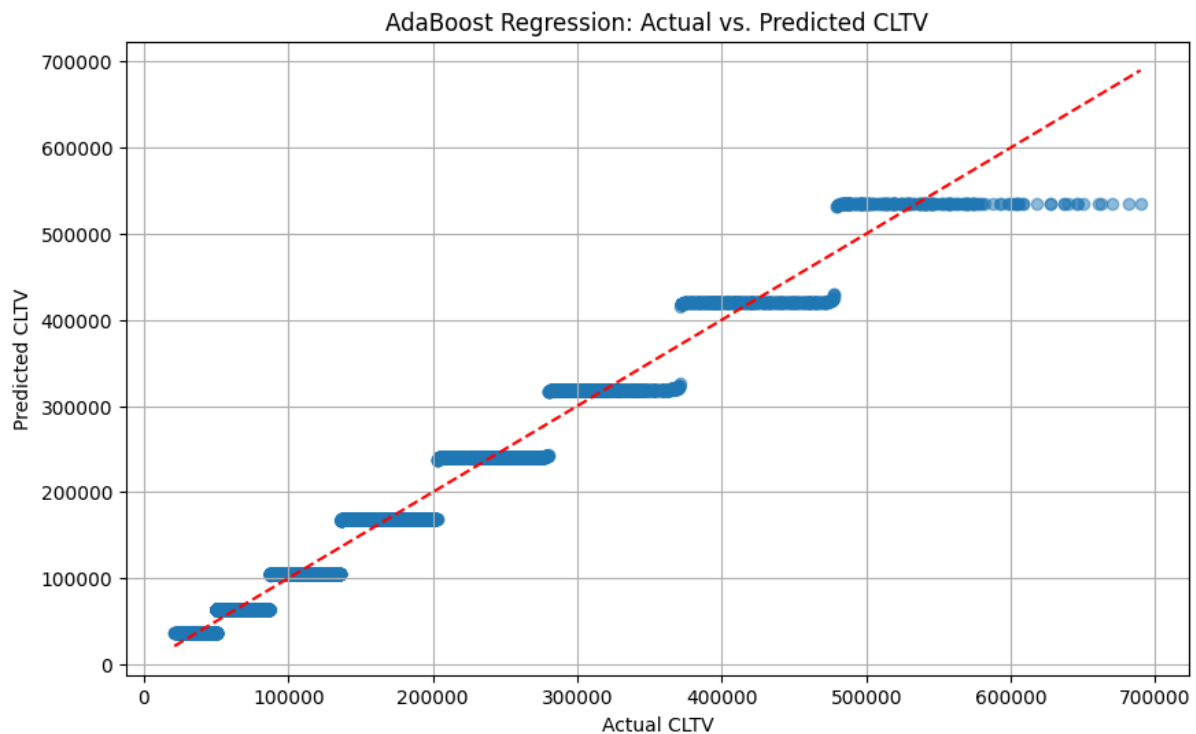


Figure 24 AdaBoost Regression: Actual vs. Predicted CLTV

(Source: Self-developed)

The Above scatterplot of AdaBoost Regression: The last one entitled Actual vs. Predicted CLTV shows the results of the model in terms of Customer Lifetime Value (CLTV). These correspond to small values most of which fall in proximity to the red diagonal line which gives indication of accurate point prediction. However, some changes may be identified because at certain levels of the actual CLTV distribution, frequency is rather high, and forecasts can be slightly underestimated or overestimated. The systematic stepwise grouping of forecasts shows that there may be further segmentation of the inputs or there are limitations to the forecast ‘boundaries’ in the model. All in all, encouraging accuracy values, especially for the test data indicating that the model may benefit from further optimization in regard to the identification of either higher end CLTV value.

4.5 XGBoost Regressor

Model	RMSE	R-Squared	MAE	MSE
XGBoost Regressor	3059.94	0.99	670.75	9363271.57

Table 5 Performance Report of XGBoost Regressor

(Source: Self-developed)

The XG Boost Regressor model of the proposed system has Moderate accuracy and resist moderate outliers and is offline by an average 3060 units from the actual values as depicted by RMSE value of 3059.94. There Is quite high value of R- square 0.99 indicates that ninety -nine per cent of the variation present in the target variable (Customer Lifetime Value or CLTV) is accounted by the model. Indeed, the MAE of 670.75 reveals the fact that the distance between the true values and the predictions of the model is only 671 units, allowing to speak about high accuracy. Total error is calculated by the so-called Mean of Squared Errors (MSE) of 9,363,271.57 for the given predictions. Thus, XGBoost is accurate model for CLTV prediction the low value of RMSE with high R-squared value and relatively low MAE indicates high precision and good prediction accuracy of the model.

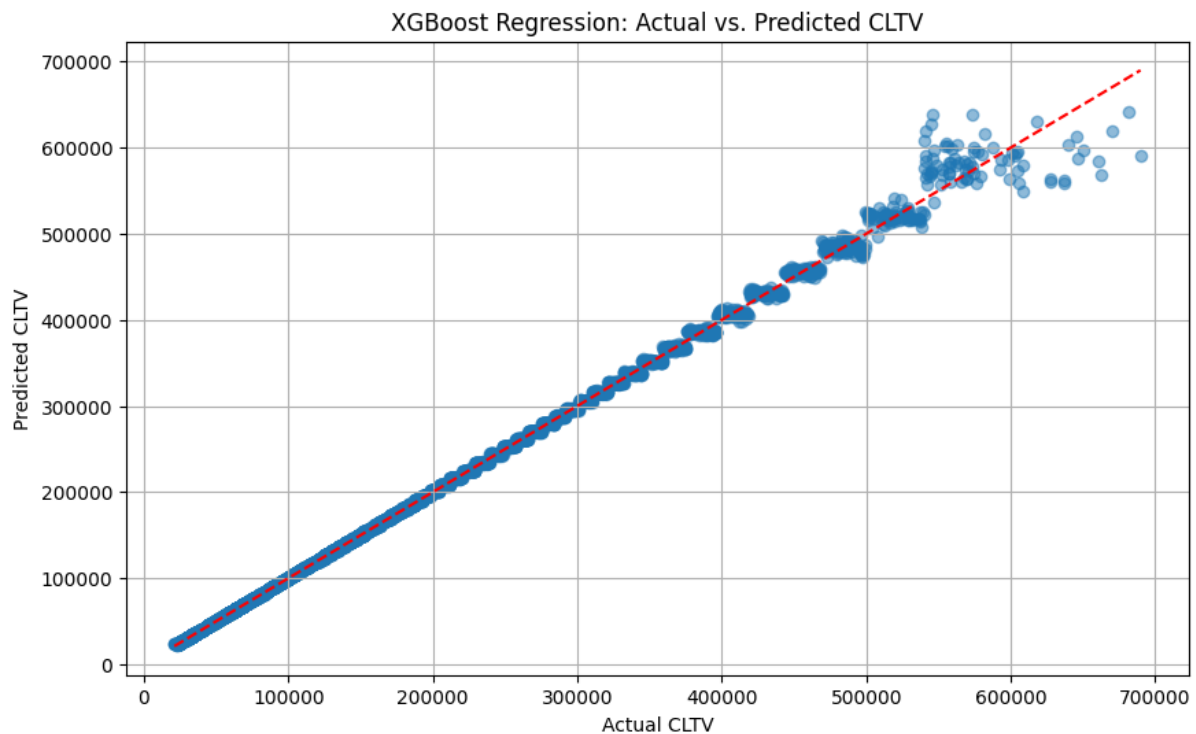


Figure 25 XGBoost Regression: Actual vs. Predicted CLTV

(Source: Self-developed)

The above scatterplot of XGBoost Regression shows the Actual vs Predicted CLTV”, the proposed model is accurate in the estimation of CLTV. Each point is as close to the red diagonal as possible, which shows that prediction is as accurate as possible for all numbers on the dataset. As it can also be observed in this figure the distribution of the points is not clustered or staircase like distribution of AdaBoost model, and the CLTV predictions continuously follow the actual CLTV values. A number of minor discrepancies take place at some extent of CLTV values; the model over or under estimate in a little calibration at the higher CLTV values. Finally, XGBoost regression’s overall performance is very high with very low errors on fluctuation in actual working range and range of Y’s variation.

Discussion

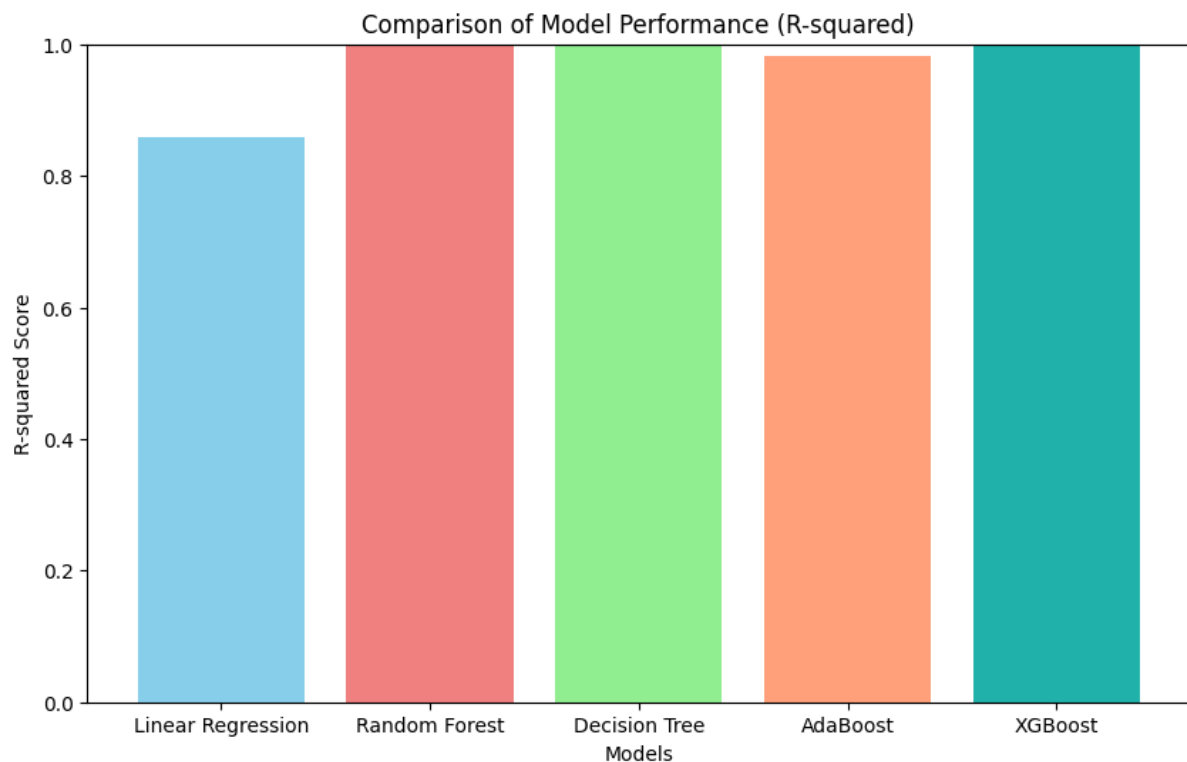


Figure 26 Comparison of Model Performance (R-squared)

(Source: Self-developed)

The bar chart indicates the performance of five models on CLTV through the use of regression models which in this case is the R-squared scores. Of all the models, Linear Regression again has slightly higher error rate and the R-squared score is less than 0.9. On the other hand, Random Forest, Decision Tree, AdaBoost, and XGBoost models, get nearly perfect scores implying high predictive ability. XGBoost performs very slightly better than other models and thus is ideal for this task. Such comparison shows how much the ensemble learning methods are beneficial in contrast to the classical linear regression in this case.

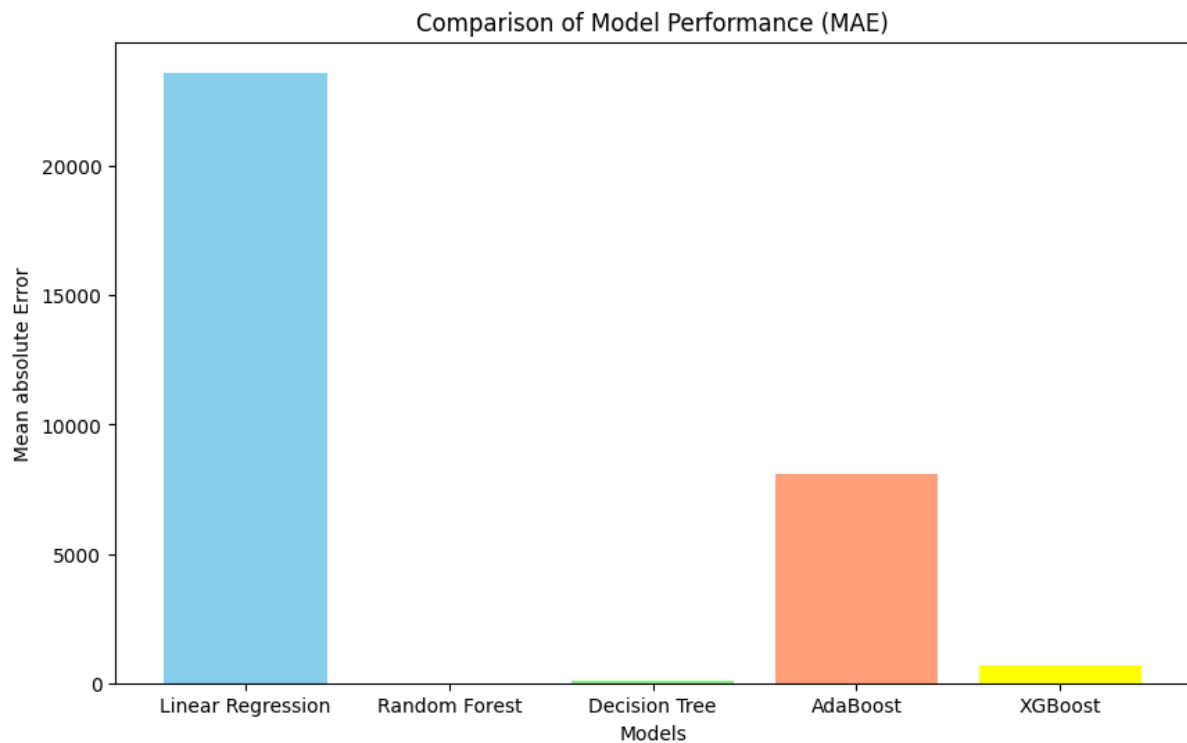


Figure 27 Comparison of Model Performance (MAE)

(Source: Self-developed)

The bar chart has been named as Comparison of Model Performance (MAE) where the mean absolute error for five regression model, which has been used to forecast the Customer Lifetime Value CLTV has been plotted. Among the models, Linear Regression has the worst accuracy, and it has the MAE of more than 20000, which is quite high and specifies an even big error rate in predicting. On the other hand, it find Random Forest, Decision Tree, AdaBoost, and XGBoost to yield much higher accuracy with much lower MAE values. Special mention is that MAE value is the lowest for XGBoost, which confirms its role of the most suitable model for this type of tasks. It also shows that while traditional linear regression has its uses, it falls short of the predictor power offered to analysts working with ensemble learning in this scenario.

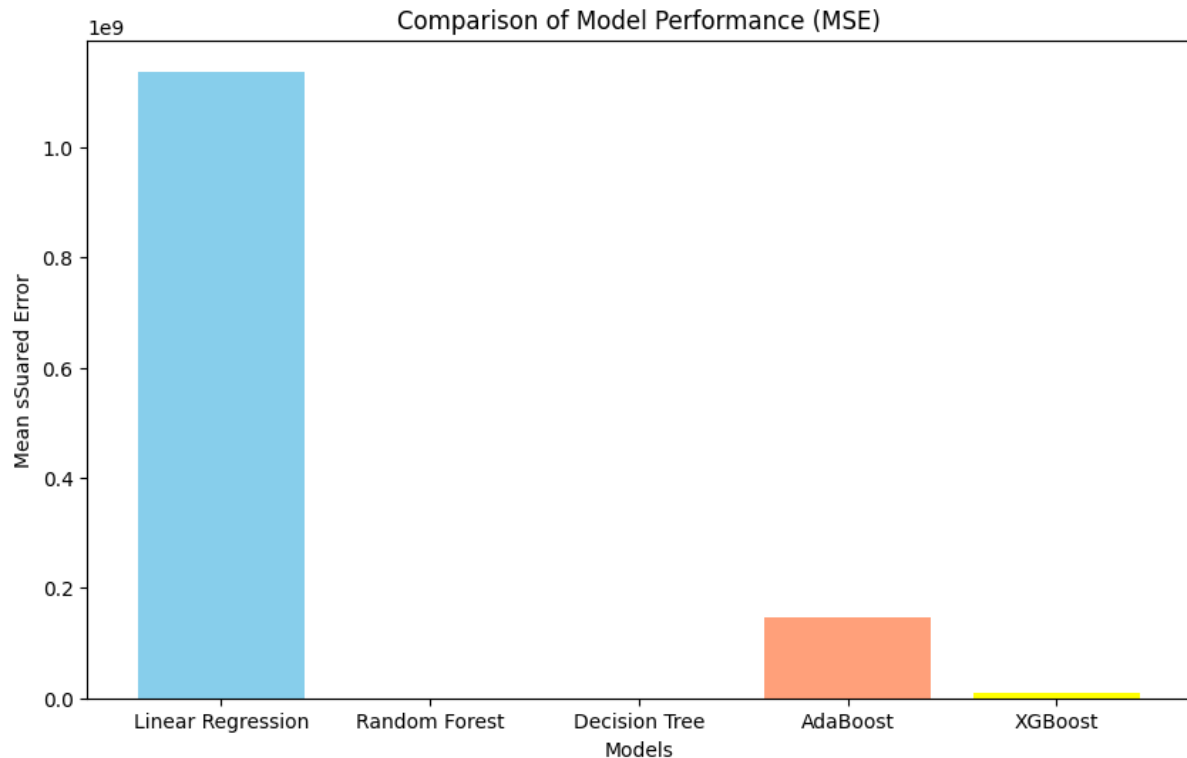


Figure 28 Comparison of Model Performance (MSE)

(Source: Self-developed)

The bar chart labelled Comparison of Model Performance (MSE) shows the MSE of five inherently used regression models for the purpose of estimating CLTV. Linear Regression gives the highest error with an MSE value more than 1 billion proving that it has worst fitness to the data. On the other hand, Random Forest, Decision Tree, AdaBoost, and XGBoost scores a lot lower MSE meaning the model is far more accurate. From the aforementioned models, the study finds XGBoost to be the most accurate with the least MSE making it highly suited for prediction purposes. Specifically, this comparison demonstrates that Linear Regression takes a very long time to predict and makes more mistakes, but combining classifiers minimizes these errors for this task.

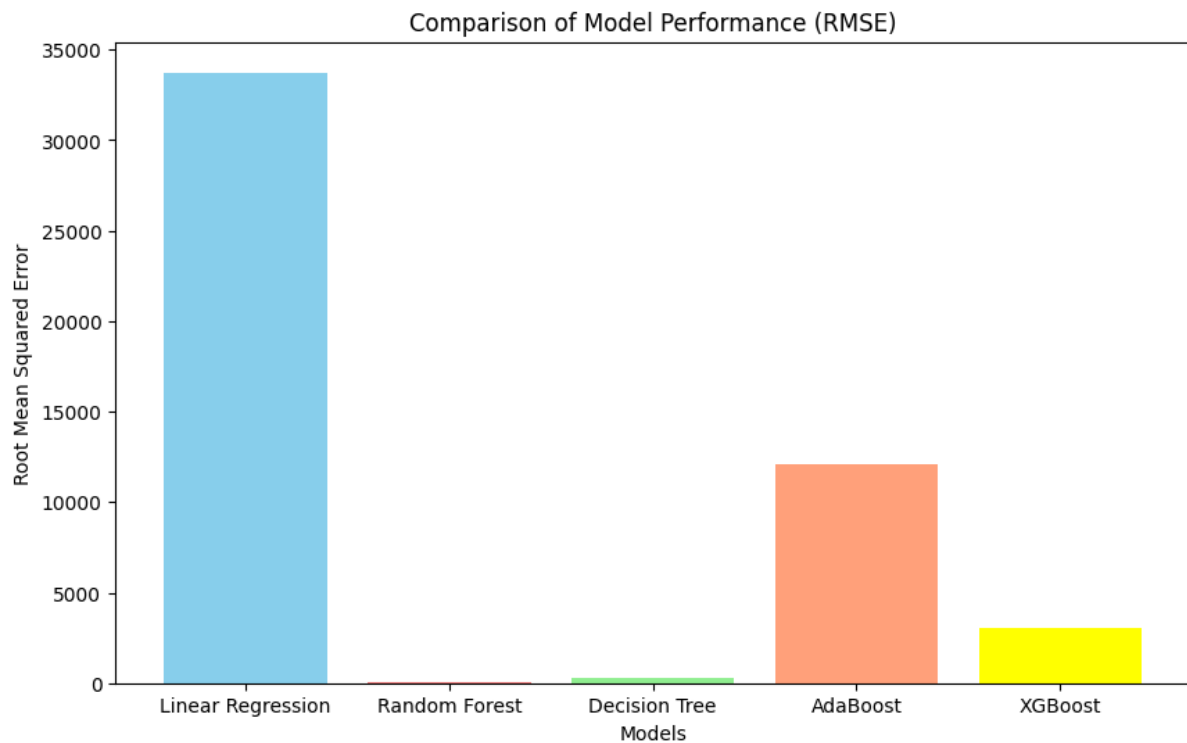


Figure 29 Comparison of Model Performance (RMSE)

(Source: Self-developed)

The above bar chart is titled Comparison of Model Performance (RMSE). From this chart, it can be easily identified that the RMSE of various models used to predict CLTV shows a significant variation. Linear Regression yields the highest RMSE, which is greater than 30,000, indicating large prediction errors. In contrast, Random Forest, Decision Tree, AdaBoost, and XGBoost models show much lower RMSE values, suggesting they are more accurate predictive models. Among these, XGBoost has the lowest RMSE, further supporting the finding that it is the most accurate model for this task. The analysis presented here demonstrates the drawbacks of the basic linear regression method and the effectiveness of new-generation ensemble methods in achieving higher accuracy in prediction results.

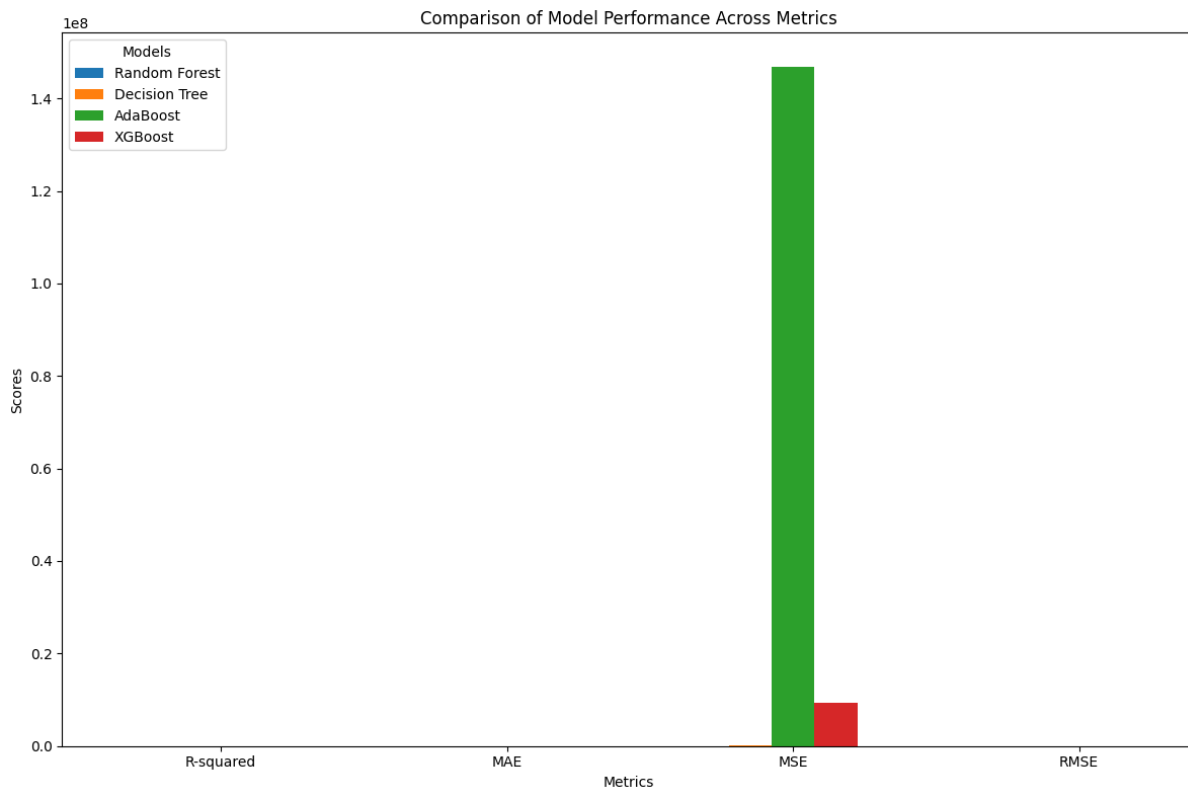


Figure 30 Comparison of Model Performance Across Metrics

(Source: Self-developed)

This bar chart analyses the results obtained from the models Random Forest, Decision Tree, AdaBoost and XGBoost on the basis of the R-squared, MAE, MSE, RMSE. In fact, AdaBoost has the highest MSE score, which is even more than double the rest of the models and metrics. This shows signs of a lapse in performance by AdaBoost for this data set. However, the other models seem to generalize better across all measures, but their individual components are obscured by the distortion AdaBoost's MSE imposes. The observed trends could be magnified further by normalizing the metrics or by using different scales to better emphasize the relative performance of each model on the different evaluation criteria.

4.6 Chapter Summary

The chapter brings a comparison of five regression models namely Linear regression, Random-forest, Decision tree regressor, Ada boost regressor and XG boost regressor to estimate the CLTV. Each model is evaluated based on key performance metrics: Mean absolute percentage error, coefficient of determination, mean absolute error, root mean square error. Linear Regression has poor results with high RMSE is 33,732.76 and MAE is 23,619.33 but it predicts 85% of the changes in the CLTV. The MSE is also quite high at 1,137,899,557.54 which evidence that this model fit poorly compared with other models. The results of model shows that residual decreases when the CLTV is low but starts to increase when CLTV raises and thus may contain non-linear relationships.

Random Forest which yields RMSE of 50.10, MAE of 5.31, and R^2 of 0.99 which confirms high accuracy with the least amount of error. Since the model can deal with nonlinear relationships it can be applied to the CLTV prediction as one can see. This scatter plot again reinforces findings that the predicted values are close to the actual values. Decision Tree Regressor is also as accurate as the previous models with an RMSE of 306.40 and an R-squared of 0.99. Still, it has a MAE of 95.52, and to the MAE comparison with other ensemble methods, it is higher so there but the optimization is possible. Nevertheless, the advantage of its capability of addressing non-linear relationships by partitioning the data into regions is useful in predicting CLTV.

AdaBoost Regressor is slightly less accurate with RMSE of 12,117.86 and R-squared of 0.98. Its MAE is slightly high revealing 8,116.48 and MSE can suggest some capability of boost in prediction seen at 146,842,709.33. XGBoost Regressor is the most accurate with RMSE of 3,059.94, close to perfect R-squared of 0.99 and a low MAE value of 670.75. From the scatter diagram, the line represents a tight fit to the actual values demonstrating it as the most accurate even for estimating the CLTV.

Chapter 5: Conclusion

5.1 Conclusion

In this study, it explored the feasibility of using machine learning techniques to estimate an important metric of business performance, Customer Lifetime Value (CLTV), using demographic and usage pattern data. The research was designed to tackle traditional CLTV estimation conundrums, specifically shortfalls in data and poor forecasts, by building relatively sophisticated models that combine customer behaviour and demographic variables. The findings indicate that the machine learning model can aid the accuracy of CLTV, helping businesses make accurate customer relationship and marketing strategy decisions.

Chapter 1 started by highlighting why CLTV is a metric for businesses to forecast a customer's revenue contribution over their lifetime. The research objectives corresponded with identifying the relations between customer engagement and CLTV, challenges in the current estimation methodologies and devising more sophisticated forecasting models. The chapter also looked at the integration of demographic and usage patterns to hypothesize that such an approach would yield more accurate predictions.

Chapter 2 presents a comprehensive literature review of existing studies that aimed to estimate the CLTV, the study model for customer behaviour, and the application of machine learning in business analytics. The chapter first identified the gaps that exist within traditional models in that they are unable to capture non-linear behaviours and real-time dynamics. This just confirmed how those advanced machine learning algorithms are needed to fill the gaps so that it can improve the prediction accuracy.

Chapter 3 In the methodology, it was described in detail that the CRISP-DM framework was used for the research process. Customer demographic and behavioural data on the dataset were fed to several regression models, including Linear Regression, Decision Tree Regression, Random Forest, AdaBoost, and XGBoost, to predict CLTV. For instance, the chapter also listed the evaluation metrics (such as RMSE, MAE, R^2) and tools (Python, Jupiter notebook) used in assessing the performance of the model.

Chapter 4: Results and analysis of the models were presented. However, huge differences across the models were found in performance. For non-linear relationships, linear regression struggled and generated high error rates. However, Random Forest, AdaBoost, and XGBoost ensemble methods outperformed, with XGBoost being the best-performing model. The

hypothesis that integrating demographic and usage patterns improves CLTV accuracy was validated.

Finally, this research illustrates how machine learning can empower CLTV estimation. The study bridges traditional gaps and offers actionable insight to give businesses powerful tools for maximizing customer value and sustainable business growth. The contributions are made to the broader field of business analytics to advance the development of future innovations in customer-centric strategies.

5.2 Recommendations

Based on the findings of this study, several recommendations can be made to improve further Customer Lifetime Value (CLTV) prediction and its application in business strategies:

Parameter Advance Machine Learning Models:

CLTV should be predicted using ensemble machine learning models like Random Forest and XGBoost, which businesses should treat as their first priority. These models, which can account for non-linear relationships and errors, showed better accuracy than traditional approaches such as linear regression. Organizations should also investigate the hybridization of algorithms to get the best of all of these.

Incorporate Real-Time Data:

To accommodate the dynamic nature of customer behaviour, the ability to integrate real-time data in systems will enable businesses to continue implementing technological advances in building real engagement. The predictions and decision-making may be refined by the incorporation of such transactional updates, behavioural trends, and environmental variables as economic indicators or market shifts.

Develop Unified Data Systems:

A centralized and unified data infrastructure is needed to process and manage all of the source data: demographic, behavioural and transactional data. This prevents inconsistency and error and improves data input quality for predictive models.

Expand Feature Sets:

Future research and applications should involve more variables, including customer feedback, social media activities and loyalty program engagements. These factors can provide further insights into customer behaviour, allowing for better prediction of CLTV.

Continuous Model Optimization:

Machine learning models can only stay relevant if updated and retrained regularly. Organizations should monitor model performance, and algorithms should be updated as they identify unfavourable customer behaviour or changes in market conditions.

Use of Insights for more customized strategies:

CLTV models should provide insights into designing personalized marketing strategies, loyalty programs and customer retention efforts. This allows businesses to concentrate on high-value customers while nurturing them.

5.3 Future Perspectives

Machine learning has several promising paths for future research and development in studying customer lifetime value (CLTV). There is a big focus on integrating real-time data streams from customer interactions like feedback systems, clickstream data and social media engagement to improve responsiveness and accuracy of prediction. Eventually, such models could take factors like macroeconomic trends, competitor activities and seasonality into account to determine customer behaviour. Further exploration should focus on building explainable machine-learning models. Ensemble methods, including XGBoost, can achieve high accuracy but are often complex and, therefore, not interpretable. The research should create transparent models that businesses can learn from the reasoning of the predictions to make better decisions.

Also, deep learning could be utilized to model complex patterns in customer behaviour. RNNs and Transformers are very good at sequential data, which could provide valuable clues about long-term customers that would be otherwise difficult to detect. Industry-specific adaptations of the CLTV models could help offer customized solutions for the retail, finance and healthcare sectors. This space is ripe for innovations through collaborative research between academia and businesses since models developed to fit with the available technologies are less likely to be actionable and will not meet the practical requirements.

5.4 Novelty of the study

The novelty of this study relies in the application of the machine learning approach has been employed for enhancing the CLTV prediction considering the demographic data along with usage pattern details. In contrast with conventional models which purely depend on past orders history besides utilizing modern data ensemble approaches such as the XGBoost to overcome

factors of non-linearities as well as dynamic customer trend. It also brings into focus the need to incorporate fresh data on the fly and add more features like customers' feedback and social media activity to make the prediction better. The work fills gaps in prior approaches by providing firms with practical recommendations for improving customer loyalty and marketing communications, marketing segmentation, and other strategic initiatives.

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